Bitcoin's Price Power Law: Stability, Maturation, and a Live Test of the Scaling Hypothesis

An empirical study of exponent stability, residual persistence, structural breaks, volatility decline, and below-trend recovery

Maxwell Koenig

Research and analytical assistance: ChatGPT (OpenAI) and Claude (Anthropic)

Technical Draft — Version 0.9 · Revised October 2026

Data through August 14, 2026 (frozen sample unchanged from v0.8)

Abstract

Bitcoin's historical price has long been noted to exhibit an unusually close power-law relationship with time. Recent work has formalized this observation, examined the stability of the fitted exponent, and proposed mechanisms linking price scaling to network adoption. This paper does not propose a new Bitcoin Power Law and does not attempt to validate a particular mechanistic derivation. Instead, we reproduce the core empirical relationship and use it as the starting point for a narrower set of stress tests motivated by Bitcoin's maturation and its 2025–2026 below-trend episode.

Using 5,873 completed UTC daily closes from July 17, 2010 through August 14, 2026, and defining time as days since Bitcoin's January 3, 2009 genesis, we estimate $\ln P(t) = \ln A + B \ln t$ with $A = 4.156832 \times 10^{-17}$, $B = 5.650938$ and $R^2 = 0.961281$. The fit is descriptively close in log–log space, but its residuals are extremely persistent, with lag-1 autocorrelation of 0.996985. Ordinary least squares (OLS) uncertainty is therefore misleading: a Newey–West estimate using a 365-day lag gives a standard error for B of approximately 0.189, versus 0.0148 under naive OLS, and a 365-day moving-block bootstrap gives a standard deviation of approximately 0.240, corroborating the widened uncertainty reported by wineLightning and Claude AI [9]. Simulation shows that even these intervals are too narrow under most of the residual models that pass a screen on Bitcoin's residual features (Section 7): under ten of twelve screened models, nominal 95% Newey–West intervals contained the true exponent in only 29–85% of simulated histories. The numerical exponent is also conditional on the genesis-date time origin. Baquero and Menezes [2] report that plausible additive shifts of the time coordinate can change the fitted exponent by nearly a factor of three, so B should be read as a within-convention estimate rather than a shift-invariant structural constant.

Rolling-window and post-2017 estimates of B vary widely: the five-year estimate fell from 4.20 to 3.26 during 2025, and the estimate from 2017 onward is 4.86. These movements follow from the short span of $\ln t$ covered by a few calendar years and from the persistence of the residuals; the 2025 fall came from 2020 observations leaving the window, not from 2025 prices. Calibrated by simulation against a declared family of persistent residual models screened on Bitcoin's residual features, these movements are compatible with a constant exponent. The principal reported statistic, the change in the local exponent implied by a quadratic fit in $\ln t$ (selected after the history had been examined, not preregistered), gives one-sided $p = 0.14$–0.35 across the twelve screened models; a secondary, multiplicity-adjusted battery of six statistics gives $p = 0.011$–0.56, below 0.05 under one model. The tests have little power, however: with critical values valid across the screened models, none detects even a three-unit decline in the exponent more than about one time in five, and inverting the principal statistic gives a 95% interval for the total decline in the local exponent of -2.5 to $+5.9$ under the best-fitting model (a negative decline would be a rise). Neither non-rejection nor the strong descriptive fit establishes that B is constant. Every screened model contains a four-year stochastic cycle that was imposed in the candidate family; that these models pass the screen is conditional on the family and the screening criteria, not independent evidence of such a cycle. All simulation results are conditional on the declared family.

A Bayesian Information Criterion (BIC) segmentation procedure is uninformative about a modern structural break once the detector is calibrated against a constant-exponent null with persistent residuals: under the v0.8 AR(1) null it selects the maximum four breaks in 99.7% of simulations and the largest real slope change sits at the 54.7th percentile, and under every screened model of the v0.9 family

it selects four breaks in all simulations. Residual unit-root tests are mixed and lag-dependent. The v0.8 Phillips–Perron statistic of -2.380 could not be reproduced and is replaced by -3.04 (11 lags) and -3.31 (automatic lag); neither rejects at residual-based 5% critical values, and no statistic rejects against a random walk with Bitcoin's declining volatility.

The current below-trend episode presents a more nuanced result. Its maximum raw log residual, -0.8165, is almost identical to the 2022 minimum and is historically mid-field by raw depth. Yet residual volatility is substantially lower in the modern market, so the same raw gap is large relative to today's smaller typical residual fluctuations. At the June 30, 2026 episode low, raw distance from the trend divided by trailing four-year residual standard deviation reached a local-sigma trend depth of -2.732; by the August 14 cutoff it had improved to -2.521. The episode ranks deepest among seven modern episodes with sufficient trailing history under that measure. A conventional locally de-meaned z-score also ranks the episode first by minimum, but only narrowly: -1.907 versus -1.847 in 2022. The distinction arises because the recent residual distribution itself is persistently centered below zero.

We further test the hypothesis that declining volatility causes slower recovery toward the Power-Law trend. The hypothesis is not supported. Lower variance mechanically makes a fixed raw gap more extreme in standardized units, but under fixed-persistence AR(1) dynamics it cannot slow first passage from an equally standardized gap. That result is conditional on a stationary AR(1) and is not evidence of mean reversion. Rolling estimates of residual persistence do not display a stable monotonic relationship with declining volatility, and historical recovery episodes are too few to identify a reliable volatility effect. Cycle highs have compressed toward the trend in every halving epoch while lows have held; against simulated shrinking-deviation nulls the asymmetry is borderline (quantile version $p = 0.015$–0.18; complete epochs only $p = 0.04$–0.28) and decisive only when the incomplete current epoch is included.

Finally, we distinguish the return required merely to keep pace with the trend from the substantially larger return required to erase an accumulated deficit. The current one-year trend hurdle is approximately 36.6%, near the 35th percentile of Bitcoin's historical rolling one-year CAGR distribution. Closing the existing gap requires approximately 199% over one year, 100.5% annualized over two years, or 74.6% annualized over three years. The current challenge is therefore less the trend's growth rate than the accumulated distance below it.

Taken together, the results do not establish that Bitcoin's historical Power Law is structurally immutable or that its exponent is constant, but neither do the completed tests provide compelling evidence of structural failure or of a structural decline in the exponent. The current episode remains a meaningful live test of the empirical scaling relationship.

1 Introduction

Bitcoin's price history is unusual even by the standards of financial time series. Its growth spans several orders of magnitude, its return distribution has changed materially as the asset has matured, and its market structure has evolved from a thin retail market into one that includes regulated derivatives, exchange-traded products, institutional custody, and sovereign and corporate ownership.

Against that changing background, a comparatively simple empirical observation has persisted: plotting Bitcoin's price against its age on logarithmic axes produces an unusually linear relationship. A common representation is

$$P(t) = A\,t^{B}, \qquad \text{or equivalently} \qquad \ln P(t) = \ln A + B \ln t, \tag{1}$$

where $P(t)$ is Bitcoin's price, t is age measured from a specified origin, A is an intercept parameter, and B is the fitted scaling exponent.

The empirical relationship has received increasing attention. Santostasi and Perrenod [1] report a fitted exponent near 5.69 and develop a mechanistic interpretation based on network adoption and generalized Metcalfe scaling. Other researchers have examined residual stationarity, exponent stability, scale invariance, and alternative structural interpretations [2, 3, 10].

Our investigation began for a less ambitious reason. While updating a Bitcoin Power Law dashboard, we observed that the fitted value of B had moved. Because B directly affects forward extrapolations, a simple engineering question became an empirical one:

Why is B changing?

That question led to several others. Is the apparent convergence of B evidence of a stable structural exponent, or would a progressively more stable estimate arise mechanically as the sample grows? Do statistical break detectors identify genuine slope changes, or merely Bitcoin's highly persistent cycle-like residual structure? Is the current below-trend episode historically unusual? Does its severity change when modern volatility is taken into account? Is the fitted model "rescuing" itself by moving downward as price underperforms? Has Bitcoin's declining volatility made recovery to trend mechanically more difficult? And, perhaps most importantly, how different is the return required to keep pace with a rising Power-Law trend from the return required to catch up after a large accumulated deficit?

Several plausible explanations did not survive testing. That is an important feature of the study rather than an inconvenience. The objective was not to demonstrate that the Power Law must hold. It was to expose the relationship to questions that could plausibly weaken it.

The resulting contribution is therefore narrower than proposing a new model. We take the existing empirical Bitcoin Power Law literature as a starting point and extend it with a set of focused diagnostics around:

1. independent corroboration of robust uncertainty in the expanding exponent estimate;
2. structural-break detection under a matched persistent-residual null;
3. systematic comparison of below-trend episodes;
4. the distinction between raw trend distance and locally standardized trend distance;
5. decomposition of current underperformance into model-refit absorption and remaining price shortfall;
6. the relationship between volatility, persistence, and recovery speed;
7. the declining growth requirement of the trend;
8. the difference between keeping pace with the trend and closing an existing deficit;
9. *(new in v0.9)* an explanation of rolling-window and post-2017 exponent estimates, and their calibration against a declared family of persistent residual models, including the statistical power of tests for a declining exponent; and
10. *(new in v0.9)* a calibrated test of whether cycle highs have compressed toward the trend faster than cycle lows.

The findings are mixed in a useful way. The historical fit remains descriptively strong. The current episode does not look unprecedented in raw distance. The recent decline in B is real numerically but not large relative to robust uncertainty. Rolling-window and post-2017 estimates move by large amounts, but movements of that size are produced routinely by a constant exponent with persistent residuals, and the available tests have too little power to settle whether the exponent is constant or declining. Structural-break tests lose their apparent significance once persistence is incorporated into the null. Declining volatility does not provide a convincing explanation for slower recovery. At the same time, the modern residual distribution has shifted materially below the long-run trend, and recent multi-year realized growth has at times lagged the contemporaneous Power-Law hurdle.

The appropriate conclusion is therefore neither "the Power Law is proven" nor "the Power Law has failed." It is that the completed stress tests do not provide compelling evidence of structural failure, while the current episode continues to test the relationship.

What changed in version 0.9. The frozen sample, the genesis-date time convention, the model specification and every v0.8 forward-monitoring threshold are unchanged. Version 0.9 adds Sections 7 and 8, revises the uncertainty, unit-root and structural-break discussion in Sections 6.2–6.4, corrects the v0.8 Phillips–Perron statistic, which could not be reproduced (Section 6.3), and updates data-availability language to the

published repository policy. Results obtained by simulation are labelled as such and are conditional on the simulated residual models; Table 17 in Section 11 classifies each principal claim by the kind of evidence that supports it. A change log accompanies this version.

2 Related Work and Positioning

2.1 The empirical Bitcoin Power Law

Santostasi and Perrenod [1] provide the recent formal account of an empirical observation that Santostasi traces to earlier informal work beginning in the prior decade. They report a regression approximately equivalent to

$$\log_{10} P(t) = -16.509 + 5.690 \log_{10} t, \tag{2}$$

with $R^2 \approx 0.961$, using data through February 18, 2026. Using our independent daily series and stopping on the same date, we obtain $B \approx 5.6912$, which is extremely close to their reported 5.690.

This reproduction is useful for a simple reason: it establishes that the empirical object examined in this paper is essentially the same price–age relationship examined in their work rather than an artifact of our dashboard implementation or data pipeline. It does not validate their mechanistic derivation. Santostasi and Perrenod relate the price exponent to network-adoption and generalized Metcalfe exponents. We do not independently estimate those network components. Our work therefore concerns the stability and behavior of the empirical price–age scaling relation, not proof of the mechanism proposed to explain it.

2.2 Stability, residuals, and competing interpretations

Recent literature has already moved well beyond simply fitting a straight line on logarithmic axes. Santostasi and Perrenod examine local slope stability and sequential convergence. Perrenod [3] separately studies residual persistence, stationarity, and declining residual amplitude: whether deviations from the trend tend to linger, whether those deviations behave as though they remain anchored to a stable long-run process, and whether their typical magnitude has shrunk over time.

Baquero and Menezes [2] provide an especially important skeptical counterpoint. Their analysis questions strong structural interpretations in three ways that matter here. First, the fitted time-domain exponent is highly sensitive to additive shifts in the assumed time origin. Second, residual and scale-invariance diagnostics used in earlier work as evidence supporting the Power Law also pass on a fitted multi-component sigmoid alternative, so those diagnostics do not uniquely identify a structural Power Law. Third, their quarterly wave-stability bootstrap rejects a Power-Law-plus-AR(1) null for Bitcoin price at $p = 0.015$, although the result is not Bonferroni-robust.

At the same time, their walk-forward analysis finds that the simple Power Law performs unusually well at 12–24 month horizons relative to flexible trend models and standard time-series baselines. That combination is important. Their result is not "the Power Law is structurally proven" or "the Power Law is useless." It is closer to: the model can remain useful as a long-horizon empirical description even when uniqueness and structural mechanism are not established. A close descriptive fit and useful forecasting properties do not by themselves imply a unique structural law.

In a subsequent working paper, Baquero and Menezes [10] ask what fifteen years of weekly prices can identify about the *shape* of the trend when deviations are as persistent as a random walk. They calibrate low-frequency statistics over a declared family of persistent deviations that includes persistence up to a random walk, a stochastic cycle of about four years, and declining volatility, and they preregister and hash their procedure before applying it. Constant drift is rejected under their family at nominal 5%, a log-age coefficient close to the published exponent is not rejected, the rejection does not survive persistence beyond a random walk, and a Gompertz curve fits about as well as the power law. They also report that tests of whether deviations revert to trend have power of at most about 0.2. Their design is the template for the

declared residual-model family used in Section 7; we adapt it to daily data and to the exponent-stability question rather than to the trend-shape question. Santostasi and Perrenod [11] have separately proposed a time-domain goodness-of-fit test in the spirit of Clauset, Shalizi and Newman [12]; we do not apply it here.

Two classical results frame all of this work. When deviations from a trend are highly persistent, regression on a function of time can produce a high R^2 and an apparently precise slope even where the true trend is different [13], and regressions on slowly varying regressors such as $\ln t$ have non-standard asymptotics [14]. Our paper therefore avoids treating R^2 as inferential evidence. With residual lag-1 autocorrelation near 0.997, high R^2 is not sufficient to establish a structural relationship.

2.3 Volatility, envelopes, and recovery

Declining Bitcoin volatility is widely documented. Perrenod [3] also reports declining amplitude specifically in deviations around the Power-Law trend. Practitioner implementations have proposed moving or empirical bands rather than treating fixed standard-deviation bands as Gaussian probability statements. Those moving-band approaches are relevant motivation, but they are not the hypothesis tested here. We do not test whether fixed bands or moving bands are preferable. Instead, we test whether the decline in residual volatility itself changes the mechanics of recovery toward the trend.

Banks [4] examines separate upper and lower Power-Law envelopes and reports materially different scaling behavior at the upper boundary, while other work on cryptocurrency return tails finds maturation-related changes and asymmetries in extreme returns [5]. Section 8 returns to the upper–lower asymmetry and tests it against simulated shrinking-deviation nulls; the descriptive observation itself is not new.

A separate line of analysis has modeled deviations from a Power-Law trend as mean-reverting processes. One 2026 practitioner study uses a rolling Ornstein–Uhlenbeck framework and argues that mean reversion may slow as Bitcoin matures [6]. We treat this as a hypothesis to be tested rather than an established result. Mean reversion here is a statistical tendency, not an assumption that price must return to trend.

A different line of work treats cyclical deviations as containing deterministic log-periodic structure rather than generic persistent noise. Wheatley et al. [7] combine a generalized-Metcalfe value model with a Log-Periodic Power Law Singularity specification for Bitcoin bubbles, while Perrenod [8] later applies log-periodic modes directly to residuals around a Bitcoin price Power Law and reports substantial residual-variance reduction. We do not test a log-periodic residual specification here. Our break-null, AR(1) mechanism and residual-model-family results should therefore be read as conditional on the persistent-stochastic residual models actually examined.

2.4 Where this study extends prior work

We do not claim novelty for exponent stability, residual persistence, declining volatility, the general idea of recovery toward trend, or the observation that cycle highs and lows behave differently. Those questions already exist in the literature.

A July 2026 study by wineLightning and Claude AI [9] independently reports materially wider autocorrelation-aware uncertainty for the time-domain exponent and also examines time-origin sensitivity. We therefore treat our robust-uncertainty result as independent corroboration rather than a novelty claim.

In our search, however, we did not identify prior work that explicitly combines the narrower analyses used here in this form, particularly:

- ranking complete below-trend episodes jointly by duration, depth, cumulative area, and modern volatility-scaled trend depth;
- quantifying how much of a live trend deficit is absorbed by regression refitting;
- explicitly separating distance from the model's trend in units of recent volatility from a conventional locally de-meaned z-score;

- testing the volatility-recovery hypothesis separately through scale, standardized depth, and persistence channels;
- distinguishing the growth required to keep pace with the current trend from the growth required to catch up to it, and benchmarking those quantities against historical rolling returns; and
- calibrating rolling-window, post-2017 and curvature exponent statistics, and the coverage of conventional intervals for B, against a screened family of persistent residual models, with explicit power calculations.

The third distinction is especially important. A trend-centered measure asks, "How far below the model are we, relative to the size of modern residual fluctuations?" A locally de-meaned z-score asks, "How unusual is this observation relative to where residuals have recently been centered?" These are related but different questions.

We therefore frame the paper as an extension and stress test of prior work rather than a new Power-Law model.

3 Data

3.1 Price series

The canonical series contains completed UTC daily Bitcoin closes from:

- July 17, 2010 through August 14, 2026;
- $n = 5{,}873$ observations;
- zero missing dates; and
- zero duplicate dates.

The current partial UTC day and all intraday spot observations are excluded from model fitting. This distinction matters because the original dashboard had previously allowed current-day information to enter paths where only completed historical observations should have been used. The production data pipeline was repaired before any research reported here was conducted. All completed studies subsequently used a byte-identical historical series, whose whole-file SHA-256 is published in the repository (see Data and code availability). Version 0.9 uses the same frozen file; no later prices enter any result.

Source attribution. Review of the production ingestion history indicates that the frozen series was stitched from two upstream sources: CryptoCompare/CoinDesk Data through January 10, 2026 and CoinGecko from January 11, 2026 through the cutoff. This attribution is inferred from the ingestion code, file paths, row counts and a precision-regime change beginning on January 11, 2026; it is not a row-by-row chain of custody. The inferred provider transition falls inside the current below-trend episode (Section 6.5). A cross-provider robustness comparison is outside the scope of this version. *Price-data credits (inferred):* CryptoCompare/CoinDesk Data (cryptocompare.com) for the earlier segment; Source: CoinGecko (coingecko.com) for January 11, 2026 onward.

3.2 Time variable

We define

$$t_i = \text{completed days since January 3, 2009.} \tag{3}$$

This is the Bitcoin-genesis convention used by the model under study. The choice is substantively important. A power law in t is not invariant to additive shifts of the time coordinate. Baquero and Menezes [2] report

that the fitted exponent changes by nearly a factor of three across their shift-parameter sweep, even though the genesis-date convention is favored by their information-criterion comparison. We therefore set t_0 to January 3, 2009 as a model convention, not as an innocuous normalization. All reported values of B are conditional on this choice. In the frozen sample t runs from 560 to 6,432 days, so the regressor $\ln t$ spans only $\ln(6432/560) = 2.44$ units.

3.3 Returns and residuals

The fitted residual is

$$r_t = \ln P_t - (\ln A + B \ln t). \tag{4}$$

Daily log returns are

$$g_t = \ln P_t - \ln P_{t-1}. \tag{5}$$

Residual volatility, return volatility, persistence, episode depth, and local-standardization measures are derived from these quantities.

4 Baseline Model

The full-history regression is

$$\ln P_t = \alpha + B \ln t + \varepsilon_t, \qquad \alpha = \ln A, \tag{6}$$

estimated by unweighted ordinary least squares (OLS) with a free intercept. The full-sample estimates are reported in Table 1.

Table 1: Full-sample estimates, July 17, 2010 – August 14, 2026 ($n = 5{,}873$).

Quantity	Estimate
A	4.156832×10^{-17}
B	5.650938
R^2	0.961281
Residual SD	0.697088
Residual lag-1 autocorrelation	0.996985

The fit is descriptively close. That statement should not be confused with a claim of conventional statistical significance or with small price-space errors. "Descriptively close" means that the fitted log–log line tracks the historical data closely across roughly six orders of magnitude of price. A residual standard deviation of 0.697 in natural-log units corresponds to a one-standard-deviation multiplicative scale of $e^{0.697} \approx 2.01$ in price. Baquero and Menezes [2], using a nearby CoinDesk window, report residual dispersion of about 0.30 in $\log_{10}$ units, or about 0.69 in natural-log units, which is closely comparable.

Statistical inference is a different question again. The residual autocorrelation is so large that the standard independent-and-identically-distributed assumptions underlying naive OLS uncertainty are badly violated. In particular, the usual tiny OLS standard errors should not be read as evidence that the exponent is known with comparable precision.

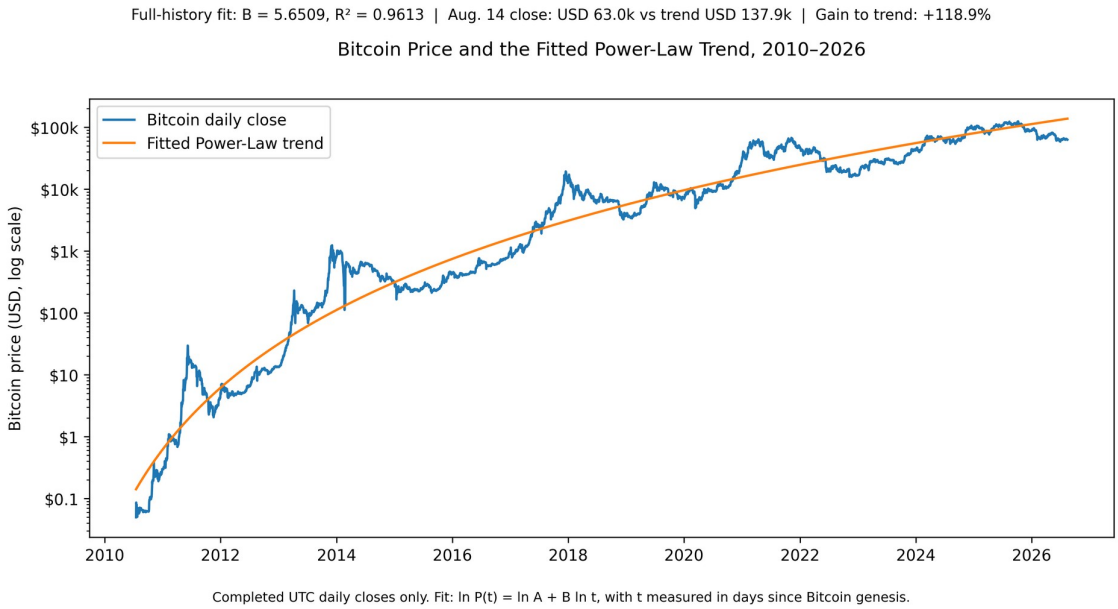


Figure 1: Bitcoin price and the fitted Power-Law trend, 2010–2026. Completed UTC daily closes on a logarithmic price axis with the full-history fit $\ln P = \ln A + B \ln t$, t in days since January 3, 2009. Rendered at publication resolution; no daily data table accompanies the figure. Inferred price sources: CryptoCompare/CoinDesk Data through January 10, 2026; Source: CoinGecko (coingecko.com) thereafter. (v0.8 Figure 1 image unchanged.)

5 Methods

Sections 5.1–5.10 retain the v0.8 methods; revisions are noted where they occur. The methods introduced in version 0.9 for exponent stability and for the compression of cycle extremes are set out with their results in Sections 7 and 8.

5.1 Sampling-frequency robustness

The regression is re-estimated using daily, weekly, and monthly observations. The resulting estimates of B differ by less than approximately 0.02. This check asks whether the headline exponent is an artifact of dense daily sampling.

5.2 Expanding-window exponent estimates

We estimate B repeatedly using expanding samples ending at successive dates. Because residual persistence makes naive standard errors unreliable, uncertainty is evaluated using:

1. Newey–West heteroskedasticity-and-autocorrelation-consistent (HAC) covariance estimates [16] across a range of lag choices; and
2. a moving-block bootstrap [17] of the $(\ln t, \ln P)$ series using several block lengths.

After this first definition, we use HAC as shorthand for heteroskedasticity-and-autocorrelation-consistent uncertainty estimates. The 365-day HAC estimate (Bartlett kernel, no finite-sample correction) is used as a practical headline value because the estimated uncertainty reaches a broad plateau around that scale.

Revision in v0.9. Whether these intervals achieve their nominal coverage is now checked by simulation: constant-B histories are drawn from each model in the declared residual-model family of Section 7.4, and we record how often the HAC and bootstrap intervals contain the true exponent (Section 6.2).

5.3 Residual stationarity diagnostics

We report:

- augmented Dickey–Fuller (ADF) tests [18] on the residuals with a constant, at fixed lags of 1, 5, 10, 21 and 33 days and with the lag chosen by AIC up to 33 (`statsmodels` 0.15.0 [27]);
- the Phillips–Perron Z_t statistic [19] with a constant and a Bartlett long-run variance at 11 lags and at the implementation's automatic lag (`arch` 8.0.0 [28], `PhillipsPerron(trend="c")`); and
- residual-based critical values for these statistics.

In plain language, these tests ask whether deviations from the fitted trend behave more like a process that remains anchored over time or like a process that can wander persistently without returning to a stable center. The ADF test is one standard way to test for this unit-root-like behavior, with the result depending in part on how much lagged information is included.

Critical values (revised in v0.9). The residual r_t comes from a regression on a constant and $\ln t$, so ordinary Dickey–Fuller constant-only critical values (-2.86 at 5%) are not appropriate: they ignore the fitted slope and over-reject. Version 0.8 also reported residual-based (Engle–Granger-style) values. Those tables [20, 21, 22] assume a stochastic $I(1)$ regressor, whereas $\ln t$ is a deterministic function of time, so they too are approximations. We therefore report three references side by side: the Phillips–Ouliaris residual-based values used by the v0.8 reproduction code (-3.96, -3.37 and -3.07 at 1%, 5% and 10%) [21]; MacKinnon's response-surface values for two variables with a constant at $n = 5{,}873$ (-3.90, -3.34, -3.05) [22]; and critical values simulated for the exact design. For the last, a random walk is simulated on the daily calendar from genesis, sampled on the observed days, added to $B\ln t$, and the identical regression and test are applied (2,000 replications), first with constant innovation variance and then with Bitcoin's observed decline in volatility (Section 6.3).

These tests should be interpreted cautiously. $\ln t$ is a deterministic function of time, not a stochastic $I(1)$ series, so conventional cointegration terminology is not a perfect match to this setting. The tests are best read as diagnostics of persistence and unit-root-like behavior around a deterministic log–log trend.

5.4 Structural-break testing

Monthly observations are segmented using exact dynamic-programming piecewise regressions with:

- minimum segment length of 12 months;
- number of breaks K selected by BIC [23]; and
- $K \in \{0,1,2,3,4\}$.

That last line simply means that the detector is allowed to choose anywhere from zero through four breaks, with BIC selecting the preferred level of complexity. The Bayesian Information Criterion balances fit against complexity: additional segments will almost always improve in-sample fit, so BIC makes each added break pay a penalty before it is retained.

The cap of four is pragmatic rather than theoretical. Four breaks already allow the algorithm to fit five separate regimes while keeping each segment long enough to be interpretable. We did not test higher caps in this study. This matters because the real-data detector selects the cap itself, so the result should not be interpreted as evidence that "four" is the true or optimal number of breaks.

The important control is not the detector's output on real data alone. We generate a constant-true-B null and run the identical detector on each simulated series. The v0.8 monthly null is calibrated to an AR(1) residual process with $\rho = 0.9160$ and residual standard deviation 0.7001, while the true monthly slope is held fixed at $B = 5.6346$. These are Monte Carlo null simulations: repeated synthetic histories in which the

true slope is constant but the residuals retain the persistence and scale of the observed monthly residual series. Version 0.9 repeats the calibration under each screened model of the declared residual-model family (Section 7.4), with daily simulation sampled at month ends.

We compare:

- selected break count;
- maximum adjacent segment-slope change; and
- total range of fitted segment slopes.

The central question is therefore a control-experiment question: would the same break detector still report apparently important breaks in a world where the exponent never actually changes? Any answer is conditional on the detector and on the simulated residual process used for the control.

5.5 Below-trend episode definition

A below-trend episode is constructed from strict crossings of the zero-residual line. The canonical episode inventory requires at least 60 days below trend; 3-day and 7-day bridge tolerances are used as robustness checks for short interruptions around zero. Under the strict rule, the sample contains ten major below-trend episodes: nine completed episodes and the current open episode. Seven of the ten have at least four years of prior residual history and can therefore be scored on the local-standardization measures in Section 5.6.

For the historical taxonomy, residuals are measured against the current full-history fitted A and B as an ex-post common yardstick. This fixed-model construction is deliberately different from the separate no-lookahead diagnostic used to ask when the current episode would have been recognized in real time.

Episodes are summarized using:

- duration;
- average residual depth;
- maximum raw residual depth;
- cumulative negative residual area; and
- modern local-standardized depth.

The current episode is included in historical rankings but is distinguished from completed episodes where recovery-time analysis is required, because the current episode has not yet completed a recovery.

5.6 Three standardized trend-depth measures

Declining residual volatility creates an interpretive problem. A raw residual of -0.8 may represent a common fluctuation in one era and an unusually large move in another. All trailing local statistics use observations strictly prior to day t, so the denominator and local mean do not incorporate the day being scored. We therefore distinguish three measures.

Trend-centered local-sigma depth.

$$D_{\sigma,t} = \frac{r_t}{s_{4y,t}}, \tag{7}$$

where $s_{4y,t}$ is the trailing four-year residual standard deviation. In plain language, this asks: how many modern-sized residual fluctuations below the Power-Law trend is today's price? A value near -2 means the price sits about two recent residual standard deviations below the model's zero line. This measures distance from the Power-Law trend's own zero line in units of recent residual fluctuation. It is not a z-score because it does not subtract the recent residual mean.

Locally de-meaned z-score.

$$z_t = \frac{r_t - \mu_{4y,t}}{s_{4y,t}}, \tag{8}$$

where $\mu_{4y,t}$ is the trailing four-year residual mean. This asks a different question: how unusual is today's residual compared with the recent regime itself? If the entire recent residual distribution has shifted below the long-run trend, the de-meaned z-score treats that shifted center as the local baseline.

RMS trend depth.

$$D_{RMS,t} = \frac{r_t}{\sqrt{E_{4y}(r^2)}}. \tag{9}$$

This retains zero as the model reference point while increasing the denominator when the recent residual distribution persistently shifts away from zero. It is therefore a middle ground between the first two measures.

The distinction is important because the recent four-year residual mean is substantially negative. The trend-centered measure asks how far Bitcoin is from the model; the de-meaned measure asks how far it is from recent experience.

5.7 Frozen-fit versus refit decomposition

To test whether the Power Law is merely "chasing" underperforming price, we freeze A and B at the beginning of the current below-trend episode and compare the resulting trend value with the trend produced by the current refit. This analysis is intentionally scoped to the current episode because it addresses a live criticism of the model: is the present trend line staying plausible simply because refitting pulls it downward as price underperforms? We are not using this calculation to rank all historical episodes.

The decline in the fitted trend is measured against the original deficit in both log space and dollar price space. Some refit drift does occur. The question is whether that drift accounts for a large or small share of the current shortfall. The results in Section 6.8 show that it accounts for only a small share.

5.8 Recovery and volatility

We test the hypothesis:

> H1: declining Bitcoin volatility causes slower recovery toward the Power-Law trend.

The hypothesis is decomposed into three mechanisms.

H1a: scale. For a fixed raw residual gap, lower residual volatility necessarily produces a larger standardized deviation.

H1b: equal-standardized-gap recovery. We examine a stationary AR(1), or first-order autoregressive, residual process

$$x_t = \rho\, x_{t-1} + \varepsilon_t. \tag{10}$$

At fixed ρ, dividing by the process standard deviation produces a standardized recursion whose first-passage law does not depend on the original scale parameter. The Monte Carlo experiment therefore illustrates rather than discovers an analytic scale-invariance result. The important distinction is between a fixed raw gap and an equally unusual standardized gap. Lower volatility can make the same raw gap more extreme, but it does not by itself make recovery from the same standardized starting point slower when persistence is unchanged.

H1c: changing persistence. The remaining possibility is that persistence itself changes as Bitcoin matures. We estimate rolling AR(1) persistence over 2-year, 4-year and 6-year windows and compare ρ with residual volatility and with calendar time. Because rolling windows overlap almost completely, these correlations are treated descriptively.

5.9 Volatility time

For completed recovery episodes, calendar recovery duration is compared with cumulative absolute return accumulated during recovery. The question is whether episodes become more homogeneous when measured in "volatility time" rather than calendar time.

5.10 Trend growth, keeping pace, and catching up

For a frozen fit,

$$P_{\text{trend}}(t) = A\,t^{B}. \tag{11}$$

The instantaneous log growth rate of the trend is

$$\frac{d\ln P_{\text{trend}}}{dt} = \frac{B}{t}, \tag{12}$$

which falls as Bitcoin ages. We calculate two distinct return requirements.

Keeping pace. The return required to maintain the current residual is

$$\frac{P_{\text{trend}}(t+h)}{P_{\text{trend}}(t)}. \tag{13}$$

Catching up. The return required for current price P_t to reach the future trend is

$$\frac{P_{\text{trend}}(t+h)}{P_t}. \tag{14}$$

The resulting annualized rates are benchmarked against historical rolling Bitcoin CAGR distributions. Rolling-window frequencies are descriptive occupancy shares, not independent probabilities.

6 Results

6.1 Baseline reproduction and sampling robustness

The full sample yields $B = 5.650938$ and $R^2 = 0.961281$. Stopping the dataset at Santostasi and Perrenod's February 18, 2026 cutoff gives $B \approx 5.6912$, almost exactly reproducing their published 5.690 on 5,696 observations. Baquero and Menezes [2] obtain approximately 5.644 on a nearby CoinDesk series ending March 25, 2026. The roughly 0.05 spread across these nearby implementations and endpoints is modest relative to the exponent itself but already larger than several naive standard-error estimates, reinforcing the need to distinguish point-estimate reproducibility from precision claims.

Daily, weekly, and monthly fits produce nearly identical exponents. These results support the claim that the empirical scaling relationship is reproducible and not sensitive to ordinary sampling frequency. They do not establish a unique structural mechanism.

6.2 The exponent is less precisely known than naive OLS suggests

The expanding-window point estimate has moved substantially over Bitcoin's history. Selected values are shown in Table 2.

Table 2: Expanding-window estimates of B (sample starting July 17, 2010).

Sample end	Expanding B
2016-12-31	5.6148
2018-12-31	5.8691
2020-12-31	5.7396
2021-12-31	5.8880
2022-12-31	5.8161
2023-12-31	5.7129
2024-12-31	5.6980
2025-12-31	5.6982
2026-08-14	5.6509

The recent decline is real numerically. From the 2021 peak to August 2026, B falls by 0.2370. However, naive OLS greatly overstates the precision of this movement. For the current full-history estimate:

- naive OLS SE: 0.0148, under the unrealistic assumption that residual observations are effectively independent;
- HAC SE with 365-day lag: 0.1892; and
- moving-block bootstrap SD with 365-day blocks: 0.2401.

The HAC-based 95% interval is approximately $[5.28, 6.02]$. The corresponding 365-day moving-block bootstrap 95% interval is approximately $[5.0466, 5.9960]$.[1]

Prior work attaches materially narrower uncertainty to related exponent estimates. Santostasi and Perrenod [1] report $\beta = 5.69 \pm 0.05$ for the global price fit and, in a different sequential Bayesian stability exercise using local scaling estimates, a final posterior approximately $N(5.729, 0.013^2)$. Those quantities arise from different procedures than the HAC and block-bootstrap uncertainty reported here and should not be treated as directly interchangeable confidence intervals. The comparison is nevertheless informative. The regression gives each day equal weight, but successive days do not provide independent evidence about B: they often reflect the same long-lived deviation from trend. Accounting for that serial dependence in the HAC and block-bootstrap calculations makes uncertainty around B much wider than naive daily OLS suggests, without changing the fitted exponent itself.

Independent work published in July 2026 reaches a closely similar conclusion by a separate route. wine-Lightning and Claude AI [9], in their Test 74 ("Honest error bars"), report a time-domain exponent of approximately 5.62 ± 0.21 after accounting for autocorrelation and summarize the widening as roughly 14 times naive uncertainty. Relative to our naive OLS SE of 0.0148, the 365-day HAC SE is about 12.8 times larger and the 365-day moving-block-bootstrap SD is about 16.2 times larger. The independently obtained magnitudes therefore closely corroborate one another despite differing implementations. We interpret the result here as independent corroboration rather than a novelty claim.

The 2021–2026 decline is therefore roughly 1.25 HAC standard errors and 0.99 365-day block-bootstrap standard deviations. More broadly, expanding-window uncertainty falls as the sample grows (Table 3).

[1] The bootstrap values are the audited v0.8 outputs. The random-number seed of the original run was not preserved; rerunning the same algorithm with the repository's publication seed (20261001) gives an SD of 0.249, which is within the variation expected between seeds.

Table 3: Expanding-window B and its 365-day HAC standard error.

End date	B	HAC SE
2016-12-31	5.6148	0.457
2020-12-31	5.7396	0.242
2024-12-31	5.6980	0.207
2026-08-14	5.6509	0.189

The observed stabilization of B is therefore at least partly the stabilization one should expect mechanically from an expanding sample. As the history grows, each new daily observation has less leverage over the full-sample slope estimate than it did when the sample was shorter.

Interpretation. The appropriate statement is not that B has stopped changing. It is:

> *The expanding estimate has become more stable, but much of that stability is consistent with increasing sample precision, and the recent decline does not by itself establish structural exponent decay.*

Convergence by itself is therefore not evidence of a fixed structural exponent. An expanding estimator becomes harder to move mechanically as the sample grows, even if the underlying residual process remains highly persistent.

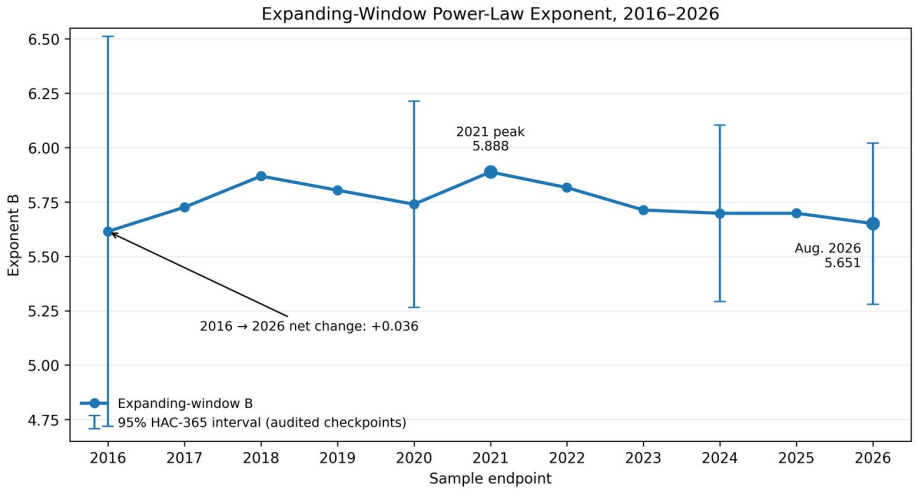


Figure 2: Expanding B estimate with robust uncertainty. Annual expanding-window estimates, 2016–2026, with 95% HAC-365 intervals at the audited checkpoints. As Figure 3 shows, these conventional intervals are themselves too narrow under most screened residual models. (v0.8 Figure 2.)

Do the conventional intervals achieve 95% coverage? (new in v0.9) The HAC and bootstrap intervals above are asymptotic devices. Their behavior at this sample length, with residuals this persistent, can be checked only by simulation. We drew 300 constant-B histories from each screened model of the declared residual-model family (Section 7.4) and from four screened-out comparison models, applied the v0.8 procedures to each history, and recorded how often the nominal 95% interval contained the true B. Monte Carlo standard errors of the coverage rates are at most 0.029. Figure 3 and Table 4 report the results.

Table 4: Coverage of nominal 95% intervals for B in simulated constant-B histories (simulation result, conditional on the residual model).

Residual model (screened)	True SD of $\hat{B}$	Mean NW-365 SE	NW-365	NW-1461	Block bootstrap
AR 0.990 + cycle 0.8 (λ=0, 1)	0.18–0.19	0.22	0.98–0.99	0.86	1.00
AR 0.995 + cycle 0.4	0.27–0.32	0.20–0.22	0.79–0.85	0.67–0.74	0.86–0.91
AR 0.997 + cycle 0.4	0.34–0.43	0.21–0.22	0.69–0.76	0.60–0.68	0.81–0.82
AR 0.998 + cycle 0.4	0.38–0.53	0.21–0.23	0.62–0.72	0.55–0.65	0.73–0.77
AR 0.999 + cycle 0.4	0.49–0.62	0.23–0.24	0.60–0.64	0.55–0.63	0.59–0.66
Fractional d=0.9 + cycle 0.8	0.57	0.24	0.62	0.49	0.75
Random walk + cycle 0.4	1.22	0.25	0.29	0.30	0.40

Ranges cover the constant-volatility ($\lambda = 0$) and shrinking-volatility ($\lambda = 1$) versions of each model where both pass the screen. NW-365 and NW-1461: Newey–West with Bartlett kernel at 365 and 1,461 lags. Block bootstrap: v0.8's 365-day moving-block percentile interval. 300 histories per model.

The 365-day Newey–West interval reaches its nominal level only under the least persistent screened model. Under the other ten screened models it contains the true exponent in 29%–85% of simulated histories. Lengthening the lag to 1,461 days does not help, because the Bartlett kernel also down-weights the longer autocovariances. The block bootstrap does somewhat better but shares the problem.

The reason is the effective sample size. When deviations persist for years, a 16-year history contains only a handful of nearly independent cycles, and a kernel estimator built from that history cannot recover the long-run variance it needs. Under the two best-fitting screened models the true sampling SD of $\hat{B}$ is 0.32–0.43, against the 0.19 reported by the 365-day HAC estimate on the real data; under a random walk with a cycle it is 1.22.

A defensible statement of precision is therefore model-indexed. Taking $\hat{B} \pm 1.96\ \mathrm{sd}(\hat{B} \mid \text{model})$, the 95% half-width for $B = 5.65$ is about 0.36 under the least persistent screened model, 0.63–0.85 under the best-fitting screened models, 1.0–1.2 under AR 0.999 with a cycle, and 2.4 under a random walk with a cycle. These are simulation results. They depend on the residual-model family and on the data-dependent screen used to select models from it, so they are sensitivity bounds rather than a single, universally valid 95% confidence interval to replace the v0.8 one. What they do indicate is that the v0.8 headline interval understates the uncertainty under most screened models rather than stating it conservatively.

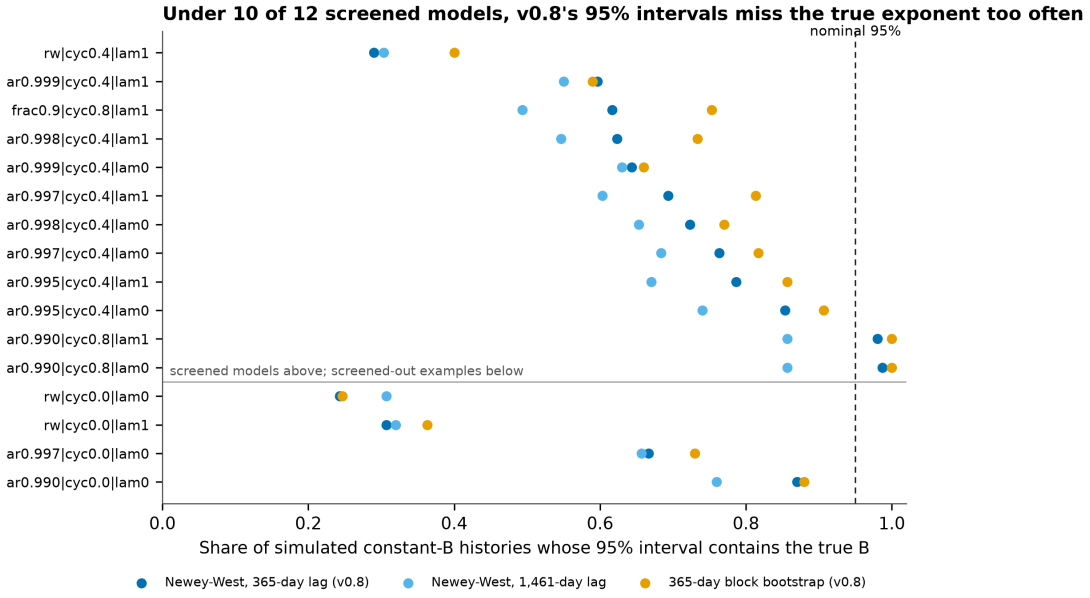


Figure 3: Coverage of nominal 95% intervals for B under the declared residual-model family (simulation). Each row is one residual model; the dashed line marks 95%. Models above the horizontal rule pass the screen of Section 7.4; those below are screened-out comparisons. 300 constant-B histories per model.

6.3 Residual persistence is extreme; stationarity evidence is mixed

The full-history residual lag-1 autocorrelation is $\rho_1 = 0.996985$. This persistence is central to almost every subsequent result. It means:

- deviations can persist for long periods;
- adjacent daily residual observations carry little independent information because today's residual is highly informative about tomorrow's;
- standard Gaussian/independence-based probability interpretations of residual bands are inappropriate given the serial dependence;
- naive regression standard errors are too small; and
- break detectors can react to long residual cycles rather than genuine slope changes.

This serial-dependence point is separate from the expanding-sample point in Section 6.2. Persistence concerns how much new information adjacent observations contain; expanding-sample stabilization concerns how much leverage any one additional observation has on the fitted exponent.

Stationarity diagnostics are not unanimous. Table 5 reports each statistic against the critical values described in Section 5.3.

Table 5: Residual unit-root statistics and critical values.

		Simulated random-walk null		5% table values	
Statistic	Observed	constant volatility 5% c.v. (p)	Bitcoin volatility 5% c.v. (p)	Phillips–Ouliaris	DF, constant only
ADF, lag 1	−2.94	−3.35 (0.15)	−4.26 (0.38)	−3.37	−2.86
ADF, lag 5	−3.17	−3.35 (0.09)	−4.22 (0.28)	−3.37	−2.86
ADF, lag 10	−3.04	−3.34 (0.12)	−4.24 (0.33)	−3.37	−2.86
ADF, lag 21	−3.43	−3.33 (0.04)	−4.13 (0.20)	−3.37	−2.86
ADF, lag 33	−3.61	−3.32 (0.02)	−4.12 (0.13)	−3.37	−2.86
ADF, AIC lag (5)	−3.17	−3.37 (0.09)	−4.15 (0.28)	−3.37	−2.86
PP Z_t, lag 11	−3.04	−3.37 (0.12)	−4.30 (0.34)	−3.37	−2.86
PP Z_t, automatic lag (34)	−3.31	−3.38 (0.06)	−4.29 (0.25)	−3.37	−2.86

Simulated critical values (c.v.) and p-values: 2,000 random walks per column, simulated on the daily calendar from genesis, added to $B\ln t$, and passed through the identical regression and test; Monte Carlo error in the 5% quantiles is about ±0.04. "Bitcoin volatility" scales the random-walk innovations by the observed annual standard deviation of daily log returns. ADF: `statsmodels` 0.15.0, constant. PP: `arch` 8.0.0, `PhillipsPerron(trend="c")`, Bartlett kernel. MacKinnon's response-surface 5% value for this case is −3.34.

Three points follow.

First, the v0.8 Phillips–Perron statistic is corrected. Version 0.8 reported $Z_t = -2.380$. The scratch implementation behind that number was not preserved, and no standard specification we tried reproduces it. On daily data, with no constant, a constant, or a constant and linear trend, at lags between 1 and 365 and at the automatic lag, the `arch` 8.0.0 statistic lies between −2.98 and −3.61; on weekly and month-end samples with a constant it lies between −2.80 and −3.43. With the v0.8 specification (constant, 11 lags) the statistic is −3.04; with the automatic lag it is −3.31. These values replace −2.380. The discrepancy with v0.8 remains unreconciled; the v0.8 conclusion that Phillips–Perron does not reject at conventional residual-based thresholds nevertheless still holds for both replacement values.

Second, the appropriate critical values are residual-based or simulated, not Dickey–Fuller constant-only values. For a random walk with constant volatility, the simulated 5% critical values for this exact design (−3.32 to −3.38) agree closely with the Phillips–Ouliaris and MacKinnon residual-based values, so those tables are a good approximation here. Against them, only the ADF statistics at 21 and 33 lags reject at 5%,

and the Phillips–Perron statistics do not. Version 0.8's statement that "several specifications reject a unit root using ordinary Dickey–Fuller critical values" is true but uses the wrong reference: at the Dickey–Fuller value of -2.86 a homoskedastic random walk around a fitted $\ln t$ trend would be rejected about 20% of the time.

Third, declining volatility changes the reference distribution. When the random walk carries Bitcoin's observed decline in volatility, the 5% critical values move to about -4.1 to -4.3, and none of the observed statistics rejects ($p = 0.13$–0.38). Equivalently, at the conventional residual-based value of -3.37 such a random walk would be rejected roughly 20% of the time. This agrees with the size distortion that Baquero and Menezes report for unit-root tests under declining volatility [10]. In the residual-model-family simulations of Section 7, the same tests also have low and erratic power against the screened stationary models (rejection rates from 2% to 54% at the residual-based 5% value), so a non-rejection is not evidence of a unit root either.

We therefore characterize the evidence as:

> *mixed and lag-dependent against a constant-volatility unit-root null, and non-rejecting against a unit-root null with Bitcoin's declining volatility; in neither case are these tests able to distinguish a highly persistent stationary residual from a unit root.*

Perrenod [3] reports rejection under all three test families on a somewhat different sample and specification. We do not interpret that difference as evidence that either analysis is necessarily "wrong." Under the present sample, construction, and lag specifications, our exact canonical series does not reproduce a clean three-test rejection. Given the deterministic nature of $\ln t$, we do not treat the disagreement as a conventional cointegration horse race. The relevant point for this paper is simpler: the residual is extraordinarily persistent and uncertainty must be handled accordingly.

6.4 Structural-break evidence is uninformative under the matched null

Applied directly to monthly Bitcoin data, the BIC segmentation selects four breaks. Taken at face value, that might appear to be strong evidence that the Power-Law exponent changes structurally. The matched null tells a different story.

Under a constant true slope with realistic persistent residuals, 299 of 300 Monte Carlo simulations (99.7%) also select four breaks. The remaining simulation selects three; none select fewer. Because the search was capped at four, these simulations do not tell us what a detector allowed to choose five or more breaks would have done. They do establish something narrower: even with a modest four-break cap, the detector almost always maxes out in synthetic data where the true exponent never changes. Thus the break count is uninformative.

The more important audited test examines break magnitude. For the real data, the maximum adjacent slope change is 13.933 and the slope range across segments is 16.585. These are differences and ranges among local segment slopes of $\ln P$ on $\ln t$, not alternative estimates of the full-history exponent. Because an allowed segment can be as short as twelve months, local slopes can sit far from the full-history $B \approx 5.65$ and can even become negative through a drawdown. The meaningful comparison is therefore with the same local-slope statistics generated by the null simulations.

Under the constant-B null, the real maximum adjacent slope change is at the 54.7th percentile and the real total slope range is at the 59.0th percentile. These percentiles are easy to interpret: roughly half of the synthetic constant-exponent histories generate slope changes at least as large as the real ones. The real-data breaks are not sitting in an extreme tail of the null distribution; they are sitting near its middle. Equivalently, $P(\text{null magnitude} \geq \text{real}) = 45.3\%$. The real break magnitudes are therefore ordinary under the null.

Calibration under the declared residual-model family (new in v0.9). The same detector was applied to 1,000 constant-B histories from each of the twelve screened models of Section 7.4. It selected the maximum

four breaks in 100% of histories under every screened model. The observed maximum adjacent slope change sits between the 38th and 60th percentiles of the twelve null distributions, and the observed slope range between the 40th and 70th. Against a declining exponent, the detector's rejection rate does not rise above its false-positive rate at any decline size (Section 7.6). The v0.8 conclusion therefore carries over unchanged to a broader set of residual models.

Interpretation. The result is not that we have proven there are no structural changes. Rather:

> *This particular structural-break detector cannot distinguish Bitcoin's observed slope changes from false breaks generated by a constant-exponent process with persistent residuals.*

Neither the number of breaks nor their size is unusual in the constant-exponent control experiment. That statement is conditional on the detector and on the simulated residual processes used as controls: the v0.8 monthly AR(1) null and the screened models of the v0.9 family.

Baquero and Menezes [2] obtain a different result with a different statistic: their quarterly wave-stability bootstrap rejects a Power-Law-plus-AR(1) null for Bitcoin price at $p = 0.015$, while noting that the result is not Bonferroni-robust. The two tests are not direct substitutes. Their statistic asks whether stable wave components are present; ours asks whether a BIC piecewise-slope detector can distinguish the observed slope changes from those generated by a constant-exponent process with persistent residuals. The divergence is itself a reason to keep the claim narrow: structural inference is detector-dependent, and our result applies to the detector and matched nulls used here.

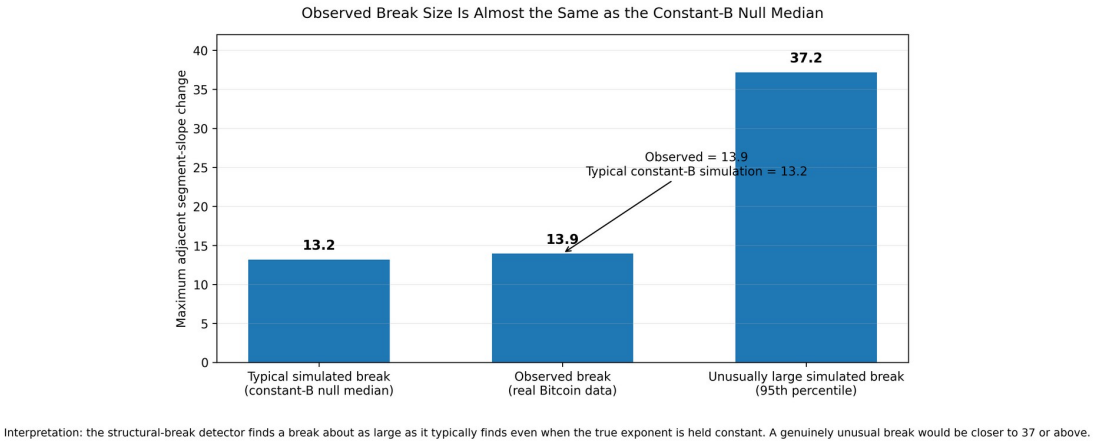


Figure 4: Structural-break magnitude versus constant-B simulations under the v0.8 monthly AR(1) null (300 simulations). The "unusually large" reference (95th percentile) is specific to that null; Section 6.4 reports the same comparison under the v0.9 residual-model family. (v0.8 Figure 3.)

6.5 The current episode is ordinary in raw depth but unusual in modern scale

The fixed-model current below-trend episode begins on November 3, 2025. A no-lookahead real-time definition identifies approximately October 28, 2025. The dates differ because they answer slightly different questions. The fixed-model date is a retrospective classification using the fully specified episode rule, while the no-lookahead date asks when the episode would have been identified using only information available at the time. The roughly one-week difference is therefore a timing-definition effect rather than a contradiction.

Through August 14, 2026, the episode has lasted 285 days under the fixed-model definition and 291 days under the real-time definition. The deepest raw residual within the current episode occurred on June 30, 2026: $r_{\min} = -0.8165$. By August 14, $r = -0.7833$. The current episode is therefore no longer at its own maximum depth.

Historically, its raw rankings are mid-field:

- duration: #4 of 10;
- average depth: #5 of 10;
- maximum depth: #6 of 10; and
- cumulative below-trend area: #5 of 10.

Nothing about the raw gap is unprecedented. The current episode is still open, however. Its duration and cumulative area are right-censored and can only increase before a future trend crossing. Their final severity ranks can therefore remain unchanged or move toward #1.

The modern-volatility comparison changes the picture. At the June 30 minimum:

- trailing four-year residual SD $= 0.2989$;
- local-sigma trend depth $= -2.732$;
- local de-meaned $z = -1.907$; and
- RMS trend depth $= -2.107$.

Among seven episodes with sufficient trailing history, the current episode ranks first by minimum on all three measures (Table 6).

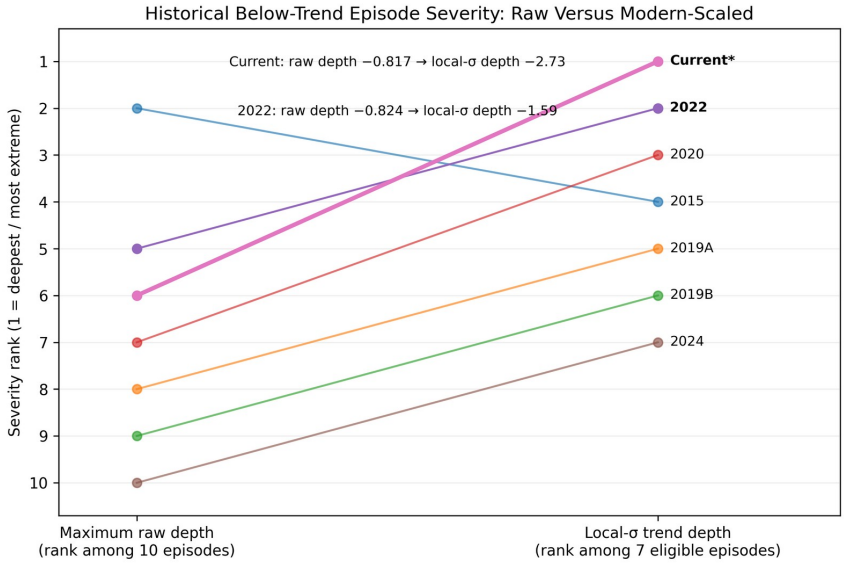
Table 6: Rank of the current episode among seven episodes with sufficient trailing history.

Metric	Rank by minimum
Local-sigma trend depth	#1 of 7
Local de-meaned z	#1 of 7
RMS trend depth	#1 of 7

The key contrast is simple: the current episode is mid-pack in raw historical distance, but it is the deepest modern episode when the same distance is scaled by today's much smaller residual fluctuations. The strength of the ranking, however, differs substantially by definition. Under the conventional locally de-meaned z-score, the current minimum is -1.907 against -1.847 in 2022: a near tie. Under the trend-centered local-sigma measure, the current value is -2.732 against -1.594 in 2022: a much larger difference.

Why? Because the trailing four-year residual distribution has itself shifted downward. The trailing mean was $+0.1294$ at the 2022 minimum, -0.2467 at the June 2026 minimum and -0.2590 by August 14, 2026. This is not a statistical nuisance to be wished away. It is part of the phenomenon.

The two measures answer different questions. The de-meaned z-score asks: *how unusual is this observation relative to where residuals have recently been centered?* The local-sigma trend depth asks: *how far below the Power-Law model's zero line is price relative to the amount residuals typically fluctuate in the modern market?* For testing the Power-Law trend itself, both are useful.



*Current episode is open/right-censored. Local-σ trend depth is raw residual divided by the strictly prior trailing-4y residual SD; it is not a de-meaned z-score.

Figure 5: Historical episode severity: raw versus local-standardized. Raw-depth ranks use all 10 episodes; local-sigma ranks use the 7 episodes with sufficient trailing history. Connecting lines identify the same episodes across different eligible sets, not a common-sample change in rank. (v0.8 Figure 4.)

6.6 The 2022 and 2026 raw gaps are essentially the same

The current episode's raw maximum depth is −0.8165. The 2022 minimum was −0.8243. The current gap is marginally shallower. If the current 2026 minimum is evaluated using the 2022 residual-volatility denominator, its trend depth becomes approximately −1.579, compared with −1.594 for 2022 itself.

Therefore the smaller modern denominator accounts for slightly more than the entire difference in trend-centered local-sigma depth: the 2026 raw gap is actually marginally shallower than the 2022 gap, so the numerator works slightly against, rather than in favor of, the larger standardized reading. An earlier internal comparison mixed different episode dates. The audited comparison above uses the matched episode minima and supersedes that preliminary result.

The correct interpretation is:

> *2022 and 2026 reached nearly identical raw distances below the trend. The current episode appears much more severe under a trend-centered local-volatility measure because residual fluctuations have become much smaller.*

6.7 The current episode deteriorated into June, but is not currently at its low

Dividing the episode into three equal time segments, the successive thirds have mean raw residuals of approximately −0.212, −0.531 and −0.694. That progression documents substantial deterioration across the episode. However, the maximum depth was reached June 30, 45 elapsed days before the end of the sample; including both endpoints gives 46 daily observations in the post-low window. Since that low, price rose approximately 7.6%, the residual improved by about 0.033 log units, and the 46-observation post-low slope is approximately flat.

Medium-horizon residual slopes remain negative: approximately −0.888 log/year over 90 days, −0.592 over 180 days, −1.017 over 365 days and −0.883 episode-wide. These horizons describe different parts of the same path. Most of the deterioration occurred before the June 30 low; the period since then has shown limited net improvement rather than a fresh collapse or a sustained recovery. The correct description is therefore:

> *The episode deteriorated steadily into the June 30 low and has not yet established a sustained recovery. It is not, however, at its worst reading as of August 14, and the post-low period is approximately flat rather than still deteriorating at the earlier pace.*

6.8 Refitting explains only a small part of the current deficit

One concern with a continuously refitted Power Law is that the model might appear stable simply because underperformance gradually pulls the fitted trend downward. We test this directly over the current episode. At episode start, $B = 5.703364$; at the end of the sample, $B = 5.650938$. The corresponding current-date trend values are approximately \$147.4k for the episode-start frozen fit and \$137.9k for the current refit. The refit lowers the trend by approximately 6.45%. Measured against the original deficit, refitting absorbs 7.84% in log space and 11.26% in price space. In dollar terms, approximately \$9.5k of the \$84.4k current-date gap measured against the frozen episode-start fit is absorbed by the refit. Thus:

> *During the current episode, the overwhelming majority of the deficit remains actual price underperformance relative to the episode-start model rather than an artifact of the model moving downward.*

There is some refit drift. The important result is that the drift is small relative to the shortfall it is sometimes accused of explaining. This statement is deliberately limited to the current episode.

6.9 Declining volatility does not provide a supported explanation for slower recovery

The original recovery hypothesis was: *lower Bitcoin volatility makes return to the Power-Law trend systematically slower.* The hypothesis sounds plausible but combines several distinct ideas.

Scale effect. From a fixed raw gap, smaller residual volatility mechanically creates a larger standardized deviation. This is true by definition.

Equal-standardized-gap effect. Suppose residuals follow $x_t = \rho x_{t-1} + \varepsilon_t$. Let s_x be the stationary standard deviation and define $u_t = x_t / s_x$. Then u_t follows the same recursion with the same ρ and standardized innovation variance $1 - \rho^2$, regardless of the original value of s_x. The scale parameter has been divided out: two stationary AR(1) processes with different volatility scales but the same persistence generate the same standardized dynamics. At fixed ρ, first-passage behavior from a starting point such as $-2s_x$ therefore does not depend on the original volatility scale.

Monte Carlo simulations illustrate the analytic result. At equal raw depth, lower-volatility regimes produce longer median crossing times. That happens because the same raw gap represents a more extreme standardized displacement when the usual fluctuations are smaller. At equal standardized depth, simulated median recovery times are approximately 398, 393 and 392 days across the three volatility regimes. Re-running the illustration with the repository's publication seed (4,000 replications, $\rho = 0.99699$, $\sigma \in \{0.85, 0.55, 0.31\}$) gives 390, 394 and 405 days, which agrees within Monte Carlo error.

The second comparison is the relevant test of the proposed scale mechanism within this conditional AR(1) framework. Once the starting deviation is equally unusual in each regime, changing the volatility scale alone does not slow recovery. The simulation should therefore be understood as a numerical illustration of scale invariance, not empirical evidence about Bitcoin. In particular, it is not evidence that Bitcoin's residuals mean-revert: Sections 6.3 and 7.4 show that the data cannot distinguish a stationary AR(1) from a unit-root residual, and under a unit root there is no finite reversion time to compare.

Persistence effect. A genuine maturation mechanism could still exist if ρ itself changed systematically as volatility declined. Rolling estimates do not show a clean pattern. Using four-year windows, era-average

ρ values remain approximately 0.9961 to 0.9974. Correlations between residual SD and ρ change sign depending on window length: positive for 2-year windows, weakly negative for 4-year windows and more negative for 6-year windows. Within individual eras, signs also change. Estimated half-lives are non-monotonic.

Across the reported four-year era-average persistence estimates, the corresponding conditional AR(1) half-lives range from roughly 177 to 266 days. Thus, small-looking changes in ρ near unity can correspond to materially different mean-reversion times. The longer four- and six-year windows lean in the H1-consistent direction: lower residual volatility is associated with higher estimated persistence, while the two-year window points the other way. Because the rolling windows overlap heavily, the sign and magnitude are not stable across reasonable window choices, and within-era signs also change, we do not treat the longer-window correlations as establishing a persistence channel.

Historical completed episodes likewise provide little identifying power, but what signal they contain does not support H1. Across the nine completed episodes, trailing residual volatility versus days to recovery has Spearman correlation of approximately $+0.14$, opposite the negative sign H1 would predict. A depth-controlled regression also produces a positive volatility coefficient, but with so few episodes the estimate is unstable; leave-one-out relationships change sign depending on which episode is omitted. Depth is more stable as an ordering variable than volatility.

The AR(1) exercise is conditional on a stationary process with $|\rho| < 1$. Because Section 6.3 finds mixed unit-root evidence, we do not claim that Bitcoin residuals are empirically established to follow a stationary AR(1). The exercise isolates one proposed mechanism: within the specified stationary AR(1) class, changing scale alone cannot change recovery from an equally standardized starting point.

Final verdict. The hypothesis that declining volatility slows Power-Law recovery is not supported. More precisely:

- the scale component is true but tautological;
- slower recovery from equal standardized depth is false under fixed-persistence AR(1); and
- a persistence-driven maturation effect is not established and cannot be cleanly excluded.

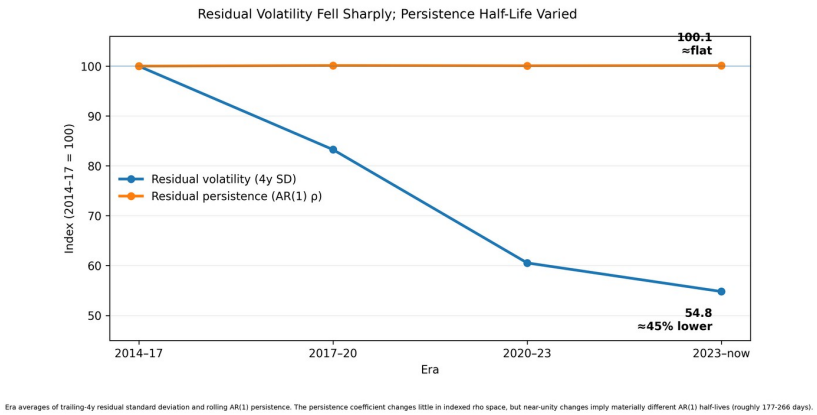


Figure 6: Residual volatility fell sharply while four-year persistence remained high but implied half-lives varied. Era averages of trailing four-year residual SD and rolling AR(1) persistence, indexed to 2014–17. (v0.8 Figure 5.)

6.10 Volatility time offers, at most, a modest secondary effect

Recovery episodes were also measured using cumulative absolute return rather than calendar time. The coefficient of variation falls from 0.884 in calendar time to 0.740 in volatility time. This suggests that part of the calendar-time variation may reflect volatility arriving at different speeds. With only nine completed episodes, we do not regard this reduction as reliably distinguishable from ordinary sampling variation. We

therefore treat volatility time as a secondary descriptive result rather than support for the main recovery hypothesis.

6.11 The trend's required growth rate is declining

For $P_{\text{trend}}(t) = At^B$, the instantaneous log growth requirement is B/t. At the end of the sample the annualized log growth is approximately 0.3209, an instantaneous arithmetic equivalent of approximately 37.84%. Exact frozen-fit CAGRs are shown in Table 7.

Table 7: Frozen-fit trend CAGR by horizon, from August 14, 2026.

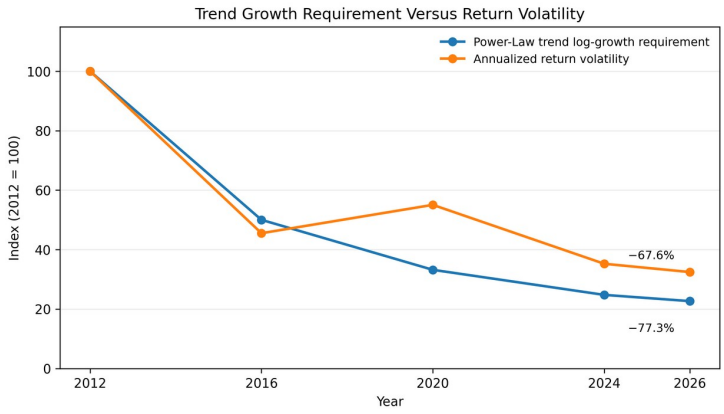
Horizon	Trend CAGR
1 year	36.63%
2 years	35.52%
3 years	34.49%
5 years	32.64%
10 years	28.93%

The required rate falls as Bitcoin ages. From 2012 to the present, the annualized trend log-growth requirement falls approximately 77.3% and annualized return-volatility SD falls approximately 67.6%. A movement-scale ratio,

$$\frac{\text{return SD}}{\text{trend log-growth requirement}}, \tag{15}$$

therefore rises from approximately 0.962 to 1.377.

This comparison should not be misunderstood. Volatility is dispersion, not expected return. It tells us how widely returns move around, not how fast their average path moves upward. An asset can be extremely volatile — very bumpy — and still fail to generate enough average upward drift to meet a growth target. The result establishes only this: the trend's required log-growth rate has fallen faster than the size of Bitcoin's typical return fluctuations. The distribution of yearly price movement is therefore not mechanically narrower relative to the trend hurdle than it was earlier in the sample. That says nothing about whether the average or realized return is high enough to clear the hurdle. Section 10 examines that growth question separately.



Selected audited checkpoints. Intermediate trend values use the reported rounded arithmetic CAGR checkpoints converted to log growth; 2012 and 2026 endpoints use the exact audited log-growth values.

Figure 7: Trend growth requirement versus return volatility, indexed to 2012. Selected audited checkpoints. (v0.8 Figure 6.)

7 Exponent Stability Revisited: Rolling Windows and Calibrated Tests

This section is new in version 0.9. After v0.8 was circulated, attention turned to two numbers that seem to point to a falling exponent: the five-year rolling estimate of B, which fell to about 3.26 by the end of 2025, and the estimate fitted only to data from 2017 onward, 4.86. This section asks why such estimates vary so much, why the five-year estimate fell during 2025, and whether either number establishes a structural decline in B. It separates three kinds of result: deterministic facts about the frozen series (Sections 7.1–7.3), simulation results that are conditional on a declared family of residual models (Sections 7.4–7.6), and the interpretation that follows (Section 7.7).

7.1 What the rolling, post-2017 and start-date estimates show

A rolling L-year estimate at date e fits Equation (6) by OLS to the observations dated in $(e-L \text{ years}, e]$, with t still measured from genesis. Table 8 reports rolling estimates for five window lengths at six checkpoints, together with the extremes and month-end standard deviation of each daily rolling path.

Table 8: Rolling-window estimates of B (deterministic, frozen series).

Window	2021-12-31	2022-12-31	2023-12-31	2024-12-31	2025-12-31	2026-08-14	Min	Max	SD
4 years	6.18	6.51	3.21	1.27	6.72	6.99	1.08	8.81	2.09
5 years	6.30	4.92	4.78	4.20	3.26	4.12	3.25	7.59	1.14
6 years	7.38	5.41	4.14	5.06	4.84	3.54	3.54	7.38	0.99
8 years	5.84	6.76	5.87	4.88	4.80	5.02	4.51	6.79	0.62
10 years	5.94	5.41	5.39	6.12	5.79	5.21	5.21	6.13	0.28

Min and Max are the extremes of the daily rolling path over all available end dates; SD is the standard deviation of the path at month ends. The five-year minimum, 3.255, occurred on 2025-12-18; the five-year maximum, 7.588, on 2020-01-12.

Three features stand out. Shorter windows swing much more: the four-year estimate ranges from 1.08 to 8.81, the ten-year estimate only from 5.21 to 6.13. The paths are not synchronized across window lengths: at the end of 2025 the four-year estimate (6.72) was above the full-history value while the five-year estimate (3.26) was far below it. And every path has risen and fallen before; the five-year estimate stood at 7.59 in January 2020.

The estimate fitted only to observations from January 1, 2017 to August 14, 2026 is 4.8645, which is 0.786 below the full-history value. Table 9 shows that this number depends heavily on the chosen start date.

Table 9: B fitted from 1 January of each year to August 14, 2026 (deterministic, frozen series).

Start	2010*	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
B	5.65	5.51	5.58	5.25	5.36	5.91	5.59	4.86	4.59	4.98	4.43	3.00	5.45

*2010 denotes the full sample, starting July 17, 2010.

Moving the start by one year changes the estimate by as much as 2.4 (between 2021 and 2022 starts), in either direction. A start in 2017 is not distinguished in the series in any way that would justify reading 4.86 as "the modern exponent": starts in 2016 and 2019 give 5.59 and 4.98.

7.2 Why short-window estimates swing so much

Write the residual relative to a true constant exponent as u_t, so that $\ln P_t = \alpha + B \ln t + u_t$. For any window W, the OLS slope is

$$\hat{B}_W = B + \frac{\sum_{t \in W} (x_t - \bar{x}_W)\, u_t}{\sum_{t \in W} (x_t - \bar{x}_W)^2}, \qquad x_t = \ln t. \tag{16}$$

The error term weights each residual by its distance from the window's mean of $\ln t$, so residuals near the ends of the window dominate, and it divides by the spread of $\ln t$ inside the window. Two properties of Bitcoin's history make that error large.

The windows are short in $\ln t$. Because $\ln t$ grows ever more slowly, a fixed calendar length covers a shrinking slice of the regressor. Ending on August 14, 2026, a four-year window spans 0.26 units of $\ln t$, a five-year window 0.33, a six-year window 0.42, an eight-year window 0.61 and a ten-year window 0.84, against 2.44 for the full sample. If the spread of x within a window is w, the denominator of Equation (16) scales with w^2; a residual pattern of a given size therefore moves a five-year slope about $2.44/0.33 \approx 7$ times as much as it moves the full-sample slope.

The residuals move together for years. With daily lag-1 autocorrelation of 0.997 and multi-year swings of a log unit or more, the residuals at the start and end of a window are not independent draws that average out. To a first approximation $\hat{B}_W - B$ is the difference between the average residual in the later and earlier parts of the window, divided by the corresponding difference in $\ln t$. A window that starts near a cycle high and ends in a below-trend phase returns a low slope; the reverse returns a high one. Both happen routinely when deviations are this persistent, whatever the true exponent.

7.3 Why the five-year estimate fell during 2025

The five-year estimate fell from 4.2004 at the end of 2024 to 3.2648 at the end of 2025, a change of -0.936. Over that year the window lost the observations of 2020 and gained those of 2025. We attribute the change day by day: on each day of 2025, the change in the estimate is split between the observation leaving the window and the observation entering it, averaging over the two orders in which the two changes can be applied, and the daily contributions are summed.

- Observations leaving the window (calendar 2020) contributed -2.03.
- Observations entering the window (calendar 2025) contributed $+1.09$.

The two single-step orders bracket these figures: removing 2020 first gives -2.93 and $+1.99$; adding 2025 first gives -1.57 and $+0.64$. All orderings agree in sign. The 2025 decline was therefore produced by the departure of 2020, not by 2025 prices, which on their own pushed the estimate up.

The mechanism is Equation (16). The full-history residual averaged -0.16 in 2020 and $+0.85$ in 2021. While 2020 was in the window, its below-trend observations sat at the low end of $\ln t$, where they steepen the fitted slope. Once 2020 left, the window began in the 2021 above-trend phase and ended in a below-trend phase, which flattens the slope. Nothing in this calculation involves a change in the long-run exponent; it is a property of where a short window starts and ends within a persistent cycle.

These three subsections describe the frozen series and are deterministic. They explain why the observed movements are large; they do not, by themselves, say whether movements of this size would be unusual if B were constant. That question requires a model of the residuals.

7.4 A declared family of residual models

To ask what a constant exponent would produce, we need simulated residual histories that resemble Bitcoin's. No single residual model is established: Section 6.3 shows that the data cannot even separate a stationary residual from a unit root. Following the declared-family approach of Baquero and Menezes [10], we therefore specify a grid of candidate models in advance, keep those that reproduce the main features of Bitcoin's residuals, and report every result across that set rather than for one preferred model.

Model. Under the null, $\ln P_t = \alpha + B \ln t + u_t$ with B constant and

$$u_t = k \left[\sqrt{1-w}\, \frac{p_t}{\mathrm{sd}(p)} + \sqrt{w}\, \frac{c_t}{\mathrm{sd}(c)} \right], \tag{17}$$

where both components are driven by Gaussian shocks scaled by $\sigma_t = h(\text{year}_t)^\lambda$, with h the observed annual standard deviation of daily log returns. The components are:

- a *persistent component* p_t, started at zero at genesis: an AR(1) with daily ρ of 0.990, 0.995, 0.997, 0.998 or 0.999 (half-lives of 69, 138, 231, 346 and 693 days), a random walk, or a fractionally integrated process with $d = 0.6$ or 0.9;
- a *stochastic cycle* $c_t = \text{Re}\, z_t$ with $z_t = \rho_c e^{i\omega} z_{t-1} + \eta_t$ [24], period $2\pi/\omega$ of four years and ρ_c chosen so that the cycle's correlation envelope halves after three periods; it starts from its stationary distribution and its peaks are *not* tied to halving dates;
- a *cycle share* $w \in \{0, 0.4, 0.8\}$ of the deviation variance; and
- a *volatility shape* $\lambda \in \{0, 1\}$: constant volatility, or the observed decline in volatility.

The constant k rescales every simulated path so that its residual standard deviation from a constant-B OLS fit equals the observed 0.697. Simulations run on the daily calendar from genesis and are sampled on the observed days. The grid has $8 \times 3 \times 2 = 48$ models; each is simulated 1,000 times with $B = 5.650938$.

Screening. Each model is compared with Bitcoin on six features of the constant-B residuals: autocorrelation at lags of 1, 365 and 1,461 days; the number of zero crossings; the ratio of residual SD in the last five years to the first five; and the share of residual variance at periods of three to five years. A model is retained ("screened") if the Mahalanobis p-value on the six features jointly is at least 0.05 and no single feature has $p < 0.01$. Two further features are reported but not screened on, because no Gaussian model in the grid matches them: the decline in daily volatility (observed 0.278, simulated about 0.31 under $\lambda = 1$) and residual skewness (observed $+0.86$, simulated about 0).

Result of the screen. Twelve of the 48 models pass (Table 10). The best fits are AR(1) with $\rho = 0.995$–0.997, cycle share 0.4 and declining volatility. Persistence is not identified: AR(1) from 0.990 to 0.999, a random walk and a fractional process all pass once a cycle is present. The 36 models that fail are kept as sensitivity extremes and reported separately.

Table 10: The twelve screened residual models and their multiplicity-adjusted p-values (simulation results, conditional on the declared family).

Persistent component	Cycle share	λ	Mahalanobis p	Smallest raw p (statistic)	Adjusted p Gaussian	Adjusted p skewed
AR 0.995	0.4	1	0.30	0.037 (mean5)	0.154	0.132
AR 0.997	0.4	1	0.29	0.076 (mean5)	0.268	0.228
AR 0.998	0.4	1	0.22	0.118 (mean5)	0.375	0.332
AR 0.999	0.4	1	0.12	0.146 (r5_2025)	0.415	0.390
Random walk	0.4	1	0.10	0.157 (r5_2025)	0.418	0.311
AR 0.997	0.4	0	0.09	0.182 (mean5)	0.480	0.473
AR 0.995	0.4	0	0.09	0.150 (mean5)	0.421	0.426
AR 0.990	0.8	0	0.07	0.028 (mean5)	0.118	0.142
AR 0.990	0.8	1	0.07	0.002 (mean5)	**0.011**	**0.006**
AR 0.998	0.4	0	0.06	0.233 (r5_2025)	0.535	0.507
Fractional d=0.9	0.8	1	0.06	0.093 (mean5)	0.299	0.296
AR 0.999	0.4	0	0.05	0.258 (r5_2025)	0.563	0.529

Ordered by goodness of fit to the six screening features. Adjusted p: Westfall–Young min-p adjustment over the six-statistic battery of Section 7.5, 1,000 histories per model. "Skewed": the same calibration with Bitcoin's right-skewed residual distribution imposed on every simulated path by rank mapping.

What the screen does and does not show about a four-year cycle. Every screened model contains the four-year stochastic cycle, and no model without it passes. Stated precisely: within the prespecified candidate family and screening criteria, models containing the imposed four-year cycle reproduced the chosen residual summaries more often than the tested non-cyclical alternatives. This is a statement about the candidate family and the screen, not independent identification of a cycle in Bitcoin. The cycle's period was fixed in advance, following Baquero and Menezes' design; two of the six screening features (the 1,461-day autocorrelation and the three-to-five-year spectral share) target multi-year periodicity directly; and no alternative structure outside the grid was tried, such as other long-memory processes, log-periodic components [7, 8] or regime-switching models. Detrending a persistent series can itself create apparent periodicity [15]. We therefore use the cycle only as a device for generating residual histories with Bitcoin-like autocorrelation, and make no claim that a four-year cycle has been discovered, validated or shown to be necessary. Testing for such a cycle independently is left to separate work.

Checks on the simulator. The simulator reproduces the benchmark results that Baquero and Menezes report for constant drift with random-walk deviations, for unit-root rejection rates and for Newey–West coverage (Appendix D). Their own low-frequency test statistic was not re-implemented, because their replication package could not be retrieved for this revision.

7.5 Calibrated tests against a constant exponent

Statistics. Each statistic below is computed identically on the real series and on every simulated history. The one-sided p-value is the share of simulated constant-B histories in which the statistic is at least as extreme, in the direction a declining exponent would produce.

- **d17**: the post-2017 estimate minus the full-history estimate (observed -0.786).
- **curv**: the implied change in the local exponent across the sample, $2c\,(x_{\text{last}} - x_{\text{first}})$, where c is the coefficient on $(\ln t)^2$ in a quadratic fit (observed -1.758). This is the paper's *principal reported test*. It was chosen after the historical data had been examined; it is not prospectively preregistered, and its p-values should be read with that in mind.
- **mean5**: the mean of the rolling five-year estimate at five checkpoints: December 31 of 2022, 2023, 2024 and 2025, and August 14, 2026 (observed 4.256). It is *not* an average over every month of 2022–2026.
- **r5_2025**: the rolling five-year estimate at December 31, 2025 (observed 3.265).
- **rmin5**: the minimum of the whole rolling five-year path at month ends (observed 3.265). Because the minimum is also taken in every simulated history, this statistic accounts for the fact that the low point was noticed after the fact.
- **exp_drop**: the fall of the expanding estimate from its post-2017 peak (observed 0.249).
- Rolling statistics for 4-, 6-, 8- and 10-year windows, the v0.8 break-detector statistics, and two variability statistics (the SD of the rolling five-year path and the range of the expanding estimate since 2017).

The first six form the *battery*. All six were examined after the data had been seen, in v0.8 and in the review that preceded this version. The smallest of the six p-values is therefore adjusted for the number of tests by referring it to its own distribution across simulated histories (the Westfall–Young min-p method [25]). The adjustment accounts for choosing among these six statistics, not for the choice of the battery itself, so even adjusted values are, if anything, too small.

Results. Table 11 reports p-values across the twelve screened models, and the critical value that controls size at 5% under all twelve.

Table 11: One-sided p-values against a constant exponent across the twelve screened models (simulation results).

Statistic	Observed	p: min – median – max	Family-wide 5% critical value	Null SD range
d17 (post-2017 minus full)	−0.79	0.039 – 0.20 – 0.32	−2.07	0.44–1.38
curv (principal reported test)	−1.76	0.137 – 0.25 – 0.35	−5.37	1.45–3.52
mean5 (five checkpoints)	4.26	0.002 – 0.13 – 0.28	2.49	0.46–1.90
r5_2025	3.26	0.066 – 0.14 – 0.26	0.27	1.57–3.27
rmin5 (selection-aware minimum)	3.26	0.38 – 0.72 – 0.81	−1.03	0.65–1.60
Rolling 10-year at cutoff	5.21	0.11 – 0.32 – 0.42	3.13	0.34–1.63
exp_drop	0.25	0.42 – 0.51 – 0.56	1.33	0.19–0.46
Secondary: battery, adjusted		0.011 – 0.40 – 0.56		

1,000 histories per model; Monte Carlo standard error of each p-value at most 0.016. "Family-wide critical value": the least favorable 5% critical value across the twelve screened models; the observed value would need to lie beyond it to reject under every screened model. Per-model adjusted p-values are in Table 10.

The observed statistics all point in the direction of a decline, but none is far into the tail of what a constant exponent produces under most screened models. The principal reported statistic, curv, gives $p = 0.14$–0.35 and does not reject under any screened model. To reject at 5% under all twelve, it would have needed to imply a decline of more than 5.4 in the local exponent; the observed value implies 1.76.

As a secondary check, the battery adjusted for multiplicity gives $p < 0.05$ under one of the twelve screened models: the least persistent model (AR 0.990), with the largest cycle share (0.8) and declining volatility, at $p = 0.011$. That value is driven by mean5, the five-checkpoint average defined above. The model fits Bitcoin's residual features no better than the others (Mahalanobis $p = 0.07$, against 0.29–0.30 for the two best fits, where adjusted p is 0.15–0.27). The range of adjusted p-values across models is a description of scenario dependence, not a single p-value for exponent decline.

Selection-aware view of the rolling minimum. A five-year estimate as low as 3.26 somewhere in the history is not unusual under any screened model ($p = 0.38$–0.81). It looks unusual only when one date, chosen after it was observed, is singled out (r5_2025, $p = 0.07$–0.26).

The observed paths are calmer than the null, not wilder. The standard deviation of the rolling five-year path and the range of the expanding estimate since 2017 are smaller than in most simulated histories (p for "at least this variable" of 0.56–0.99 and 0.89–0.99). Either the family slightly overstates low-frequency variability, which would make the p-values above somewhat conservative, or Bitcoin sits at the less persistent end of the family, which is where the smaller p-values arise. One history cannot separate the two.

Sensitivity. Re-simulating each screened model with volatility shape $\lambda = 0.5$ and 1.25 and with cycle periods of 3.5 and 4.5 years gives 38 distinct variants, of which 26 pass the same screen. Across all 38 screened models (12 base, 26 variants), the adjusted p-value ranges from 0.000 to 0.56. Three are below 0.05, and all three are versions of the AR 0.990, cycle-share 0.8, declining-volatility model. The eight best-fitting of the 38 give 0.12–0.34. Imposing Bitcoin's right-skewed residual distribution on every simulated path changes the adjusted values only slightly (0.006–0.53; Table 10).

Figures. Figures 8–10 show the observed paths against bands generated by a constant exponent under the screened models. Each band is the envelope of the twelve models' pointwise 95% ranges (at each date, the lowest 2.5th and the highest 97.5th percentile across models), so it is wider than the range of any single model. Every rolling path, at every window length, stays inside this envelope, as does the start-date curve. The expanding estimates leave it only in the first months after the 2011 and 2017 peaks, and only upward.

Staying inside the envelope does not mean staying inside the range of each screened model. Pointwise bands are a visual aid; the tests above are the formal comparison. Figure 11 shows the p-values by model.

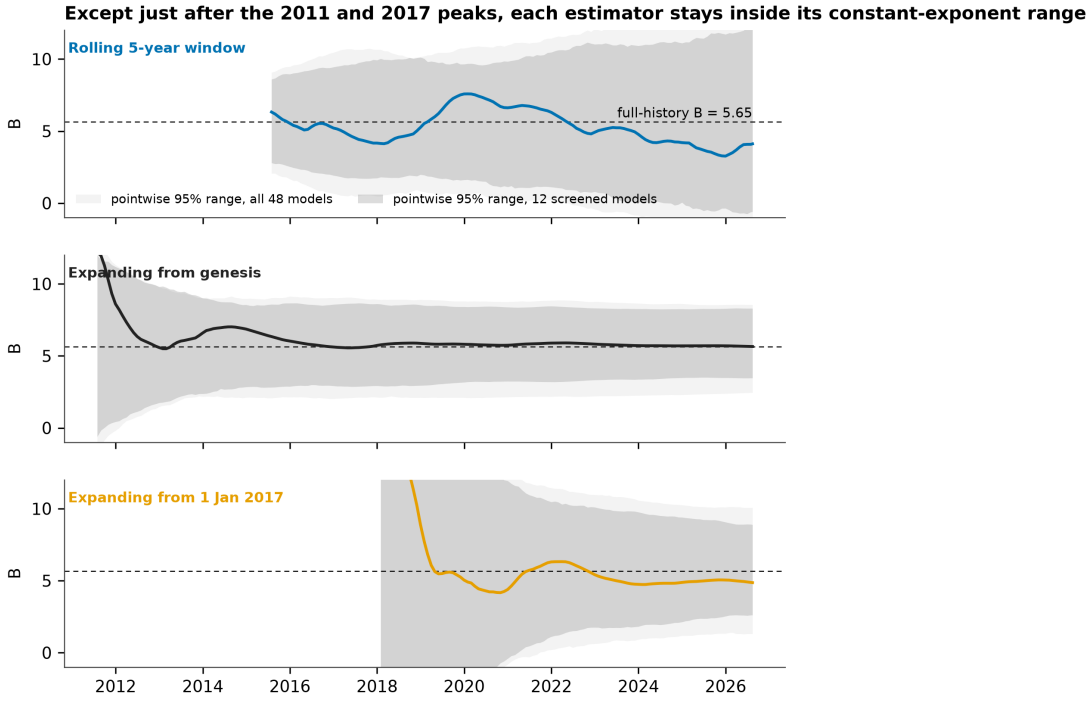


Figure 8: Rolling five-year, expanding and post-2017 expanding estimates of B against bands produced by a constant exponent (simulation). Each band is the envelope of the models' pointwise 95% ranges: at each date, the lowest 2.5th and the highest 97.5th percentile across models. Dark band: the twelve screened residual models; light band: all 48 models. Observed paths are drawn at month-end resolution.

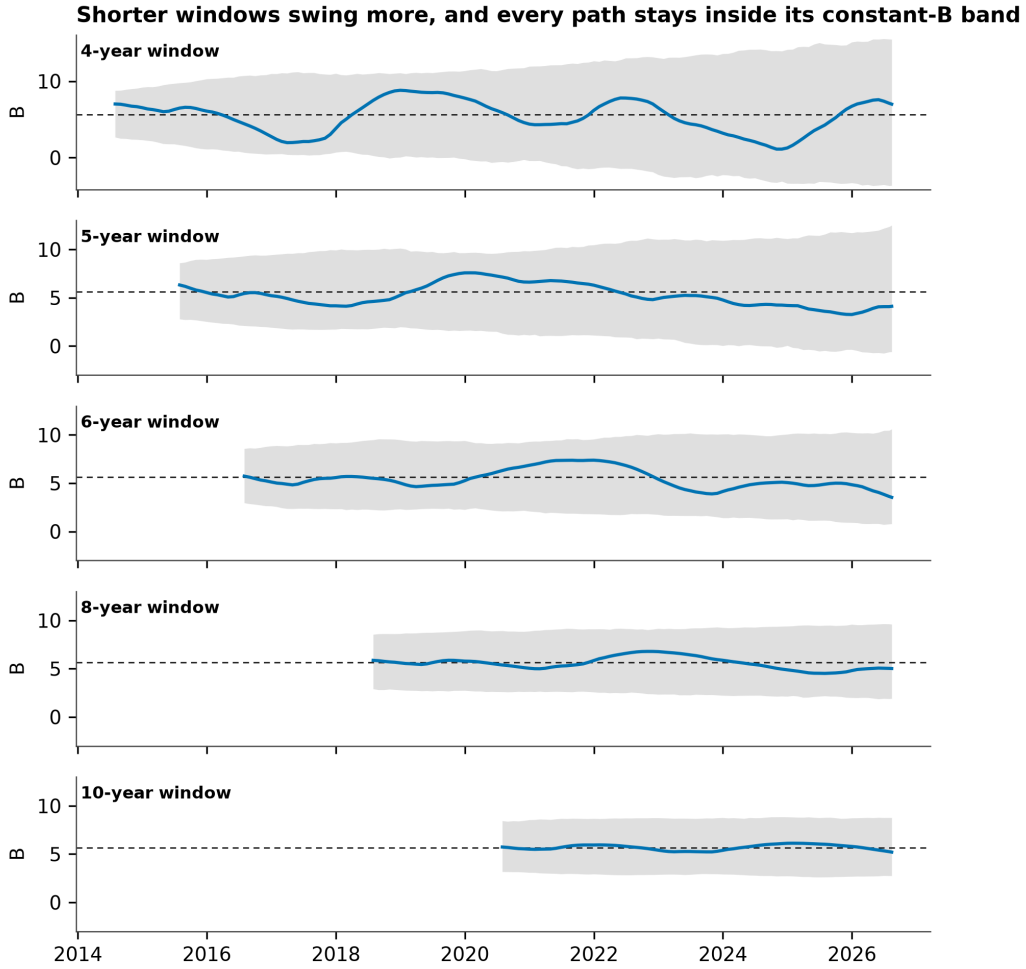


Figure 9: Rolling estimates of B for 4-, 5-, 6-, 8- and 10-year windows against the envelope of the twelve screened models' constant-B pointwise 95% ranges (simulation). Dashed line: full-history B.

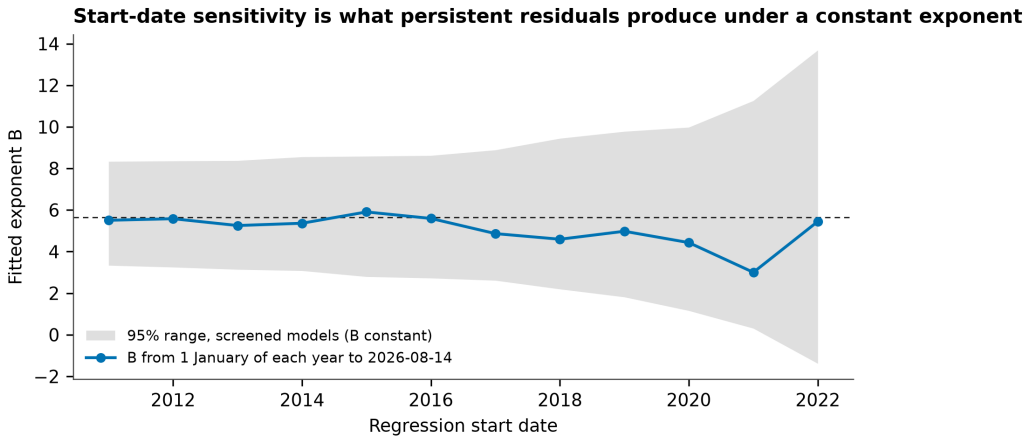


Figure 10: B fitted from 1 January of each year to August 14, 2026, against the envelope of the twelve screened models' constant-B pointwise 95% ranges (simulation).

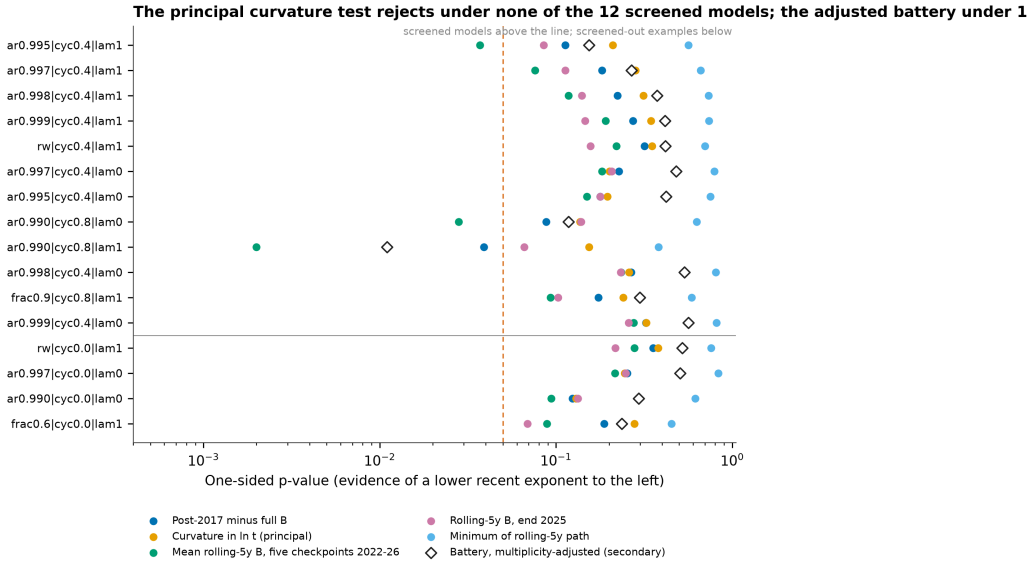


Figure 11: One-sided p-values against a constant exponent, by residual model (simulation). Dots: individual statistics ("mean rolling-5y B" is the mean of the five checkpoints defined in Section 7.5). Open diamonds: the battery after Westfall–Young adjustment. Models above the horizontal rule pass the screen; examples below do not. Dashed line: $p = 0.05$.

7.6 Statistical power against a declining exponent

A test that rarely rejects a false null cannot support a claim of constancy. We therefore measure how often each statistic would detect a declining exponent of known size.

Alternatives. Let $x = \ln t$, with x_0 and x_1 its first and last observed values. Histories are generated as $\ln P_t = \alpha + Bx_t + g_D(x_t) + u_t$, with u_t drawn from a screened model, for two shapes of decline:

- *gradual decline across the sample*: the local exponent $\beta(x) = B + D\left(\frac{1}{2} - \frac{x-x_0}{x_1-x_0}\right)$ falls linearly in $\ln t$ from $B + D/2$ to $B - D/2$, so

$$g_D(x) = D\left[\frac{x-x_0}{2} - \frac{(x-x_0)^2}{2(x_1-x_0)}\right]; \tag{18}$$

- *decline after 2017*: $\beta(x) = B$ before $x_k = \ln t_{\text{2017-01-01}}$ and falls linearly to $B - D$ at x_1, so

$$g_D(x) = -D\,\frac{(x-x_k)^2}{2(x_1-x_k)} \quad \text{for } x \geq x_k, \qquad 0 \text{ otherwise.} \tag{19}$$

D is the total fall in the local exponent and takes the values 0, 0.25, 0.5, 1, 1.5, 2 and 3. For each screened model, 1,000 residual paths are drawn once, with seeds independent of the calibration draws, and reused for every D, so that the $D = 0$ case gives an out-of-sample false-positive rate and differences across D are not blurred by simulation noise. Monte Carlo standard errors are at most 0.016.

Critical values. Two kinds are used. A *family-wide* critical value is the least favorable 5% critical value across the twelve screened models; it is the only kind that controls the false-positive rate when, as here, the true residual model is unknown. A *model-specific* critical value uses the 5% point of the model that generated the noise. It is an oracle benchmark: an upper bound on power, attainable only if the residual model were known.

False-positive rates. With family-wide critical values, the out-of-sample false-positive rate averages 0.006–0.023 across statistics, with a maximum of 0.063 for any single model (the excess over 0.05 reflects Monte Carlo error in the least-favorable quantile). With model-specific critical values it averages 0.045–0.059, close to nominal.

Power. Table 12 and Figure 12 report rejection rates averaged over the twelve screened models.

Table 12: Rejection rates at the 5% level against a declining local exponent, mean over the twelve screened models (simulation).

	Gradual decline, total D			Decline after 2017, total D		
Test	1	2	3	1	2	3
Family-wide critical value						
Curvature (principal reported test)	0.04	0.09	0.18	0.04	0.08	0.14
Post-2017 minus full	0.03	0.05	0.09	0.04	0.09	0.16
Mean rolling-5y (five checkpoints)	0.02	0.04	0.06	0.03	0.07	0.15
Rolling 5-year at cutoff	0.02	0.03	0.04	0.03	0.05	0.09
Rolling 8-year at cutoff	0.02	0.04	0.07	0.03	0.08	0.18
Expanding-B drop from peak	0.04	0.08	0.14	0.03	0.06	0.10
Break detector (largest change)	0.02	0.02	0.02	0.02	0.02	0.02
Model-specific (oracle) critical value						
Curvature	0.13	0.26	0.40	0.12	0.22	0.35
Post-2017 minus full	0.11	0.18	0.28	0.14	0.27	0.42
Mean rolling-5y (five checkpoints)	0.10	0.19	0.30	0.16	0.35	0.53
Rolling 5-year at cutoff	0.08	0.11	0.15	0.11	0.19	0.30
Rolling 10-year at cutoff	0.13	0.26	0.41	0.18	0.41	0.60
Battery, multiplicity-adjusted	0.14	0.29	0.45	0.17	0.36	0.54
Break detector (largest change)	0.06	0.06	0.06	0.06	0.06	0.06

1,000 histories per model and value of D, common random numbers across D. Across individual screened models the battery's oracle power at $D = 3$ ranges from 0.32 to 0.84 (gradual) and from 0.37 to 0.98 (after 2017). Full per-model results are in the reproducibility package (`power_curves.csv`).

With critical values that are valid across the screened family, no statistic detects even a three-unit decline in the exponent more than about one time in five. With the residual model known, the best statistics reach 0.45–0.60 at $D = 3$, and rolling five-year statistics reach only 0.15–0.30. The v0.8 break detector has no power at all: its rejection rate does not rise with D.

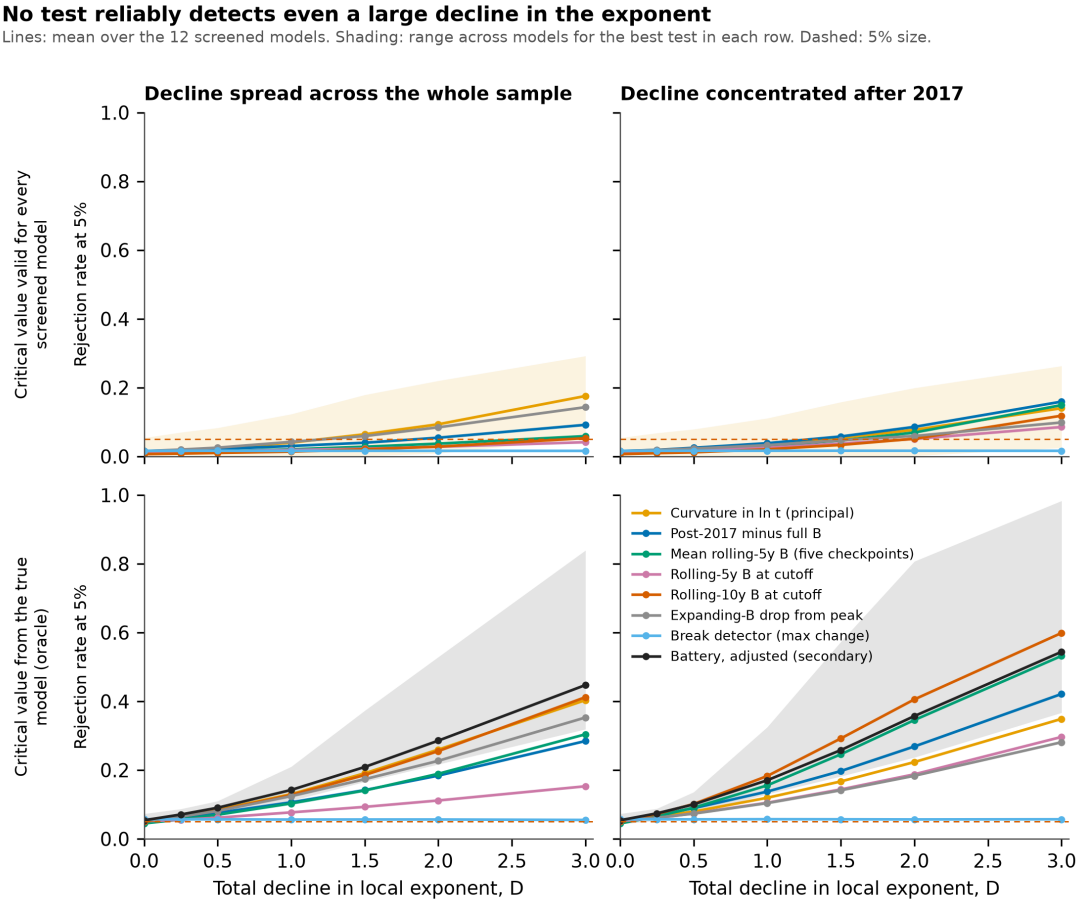


Figure 12: Power against a declining local exponent (simulation). Top row: critical values valid for every screened model; bottom row: critical value of the model that generated the noise (oracle). Left: gradual decline across the sample; right: decline after 2017. Lines are means over the twelve screened models; shading shows the range across models for the best test in each row; dashed line, 5%.

How large a decline is compatible with the data? Under the gradual-decline alternative the curvature statistic estimates $-D$ without bias, because g_D is exactly quadratic in $\ln t$. Its null distribution therefore inverts directly to a confidence interval for D: $D \in [q_{0.025} - \text{obs},\ q_{0.975} - \text{obs}]$, where q are quantiles of the statistic under the constant-B null. The point estimate is $D = 1.76$. The 95% interval is $[-2.5, 5.9]$ under the best-fitting screened model; it narrows to $[-1.0, 4.5]$ under the least persistent screened model and widens to $[-4.2, 7.8]$ under a random walk with a cycle. Every interval contains zero, and every interval also contains declines of four units or more. Taken at face value, the data are compatible both with an exponent that has stayed constant and with one that has fallen from about 8 to about 3.5 over 2010–2026.

7.7 Interpretation

The movements in the rolling and post-2017 estimates are real features of the frozen series, and Sections 7.2–7.3 explain them without reference to any change in the long-run exponent. Under the screened residual models, a constant exponent routinely produces movements of the same size. Only one corner of the candidate family rejects a constant exponent, and it is not the best-fitting corner.

That is not evidence that B is constant. The same tests detect large declines only rarely, and a confidence interval inverted from the principal reported test spans both no change and a large decline. Nor does the strong descriptive fit of Section 4 establish constancy: with residuals this persistent, a high R^2 is compatible with very different trends [13]. The defensible conclusion is narrower:

Neither the rolling-window movements nor the post-2017 estimate establish a structural decline

in B, and the available tests cannot establish that B is constant. Within the screened residual-model family, the question is not answerable with sixteen years of data. Every probability statement in this section is conditional on that family.

8 Asymmetric Compression of Cycle Extremes

This section is new in version 0.9. Declining residual amplitude around the Power-Law trend has been reported before: Perrenod [3] reports that the average absolute residual has fallen by about 60% and fits separate declines to the positive and negative residuals, without comparing their rates. Separate upper and lower envelopes have also been reported to scale differently [4]. Here we ask a narrower question: have the highs of the residual cycles compressed toward the trend faster than the lows, and is that asymmetry distinguishable from the symmetric or skew-driven shrinkage that a persistent residual with declining volatility would produce anyway? Differences between the upper and lower envelopes have been noted before [4]; the contribution here is the calibration.

8.1 What is observed

We group the full-history residuals by the intervals between Bitcoin's block-reward halvings: E0 (from the start of the sample on July 17, 2010, to November 27, 2012; partial), E1 to E3 (complete), and E4 (from April 20, 2024, to the cutoff; incomplete). The halving dates are used only to partition the sample into four-year blocks; nothing in this section tests whether halvings cause cycles, and the simulated residuals are not tied to halving dates.

Table 13: Largest and smallest full-history residual in each halving epoch (ln units, deterministic).

Epoch	Coverage	Maximum	Minimum
E0	partial (sample starts July 17, 2010)	2.76	−1.58
E1	complete	2.50	−0.97
E2	complete	1.86	−0.83
E3	complete	1.27	−0.82
E4	incomplete (from April 20, 2024)	0.30	−0.82

Epoch maxima have fallen in every epoch, while minima have been nearly unchanged since E2. Quantile regressions [26] of $\ln P$ on $\ln t$ show the same pattern across the whole distribution: the slope at the 95th percentile (4.93) is 0.92 below the median slope (5.85), while the slope at the 5th percentile (5.88) is 0.04 above it. The upper envelope has flattened and the lower envelope has not.

8.2 Tests against shrinking-deviation nulls

Statistics. With b_τ the quantile-regression slope at quantile τ:

$$A_q = b_{0.95} + b_{0.05} - 2\,b_{0.50}, \tag{20}$$

which is close to zero when the residual distribution shrinks symmetrically and negative when only the top compresses (observed −0.885). With $s_{\max}$ and $s_{\min}$ the least-squares slopes of the epoch maxima and minima across epochs,

$$A_{ep} = s_{\max} + s_{\min}, \tag{21}$$

computed for the three complete epochs E1–E3 (primary; observed −0.544), for E0–E3 (−0.268), and for E0–E4 (−0.447), which adds the incomplete current epoch.

Nulls. For each screened model of Section 7.4, 400 constant-B histories are drawn under two nulls. The *symmetric shrinking null* uses the Gaussian paths as simulated; under declining volatility they shrink. The *skewed shrinking null* maps each path onto Bitcoin's observed distribution of residuals standardized by a rolling four-year SD (skewness $+0.47$), keeping the path's timing and persistence. Under this null, highs compress more than lows in absolute terms purely because of the right skew. It is the harder benchmark. The Monte Carlo standard error of each p-value is at most 0.025, and p-values below $1/400$ cannot be resolved.

Table 14: One-sided p-values for top-only compression, range across the twelve screened models (simulation results).

Statistic	Symmetric shrinking null	Skewed shrinking null
A_q, quantile asymmetry	0.015 – 0.135	0.033 – 0.178
Lower-quantile slope alone ($b_{0.05} - b_{0.50}$)	0.09 – 0.62	0.11 – 0.58
A_{ep}, complete epochs E1–E3	0.040 – 0.24	0.08 – 0.28
A_{ep}, E0–E3	0.048 – 0.15	0.11 – 0.23
A_{ep}, E0–E4 (incomplete E4 included)	< 0.0025	0.000 – 0.010

400 histories per model and null.

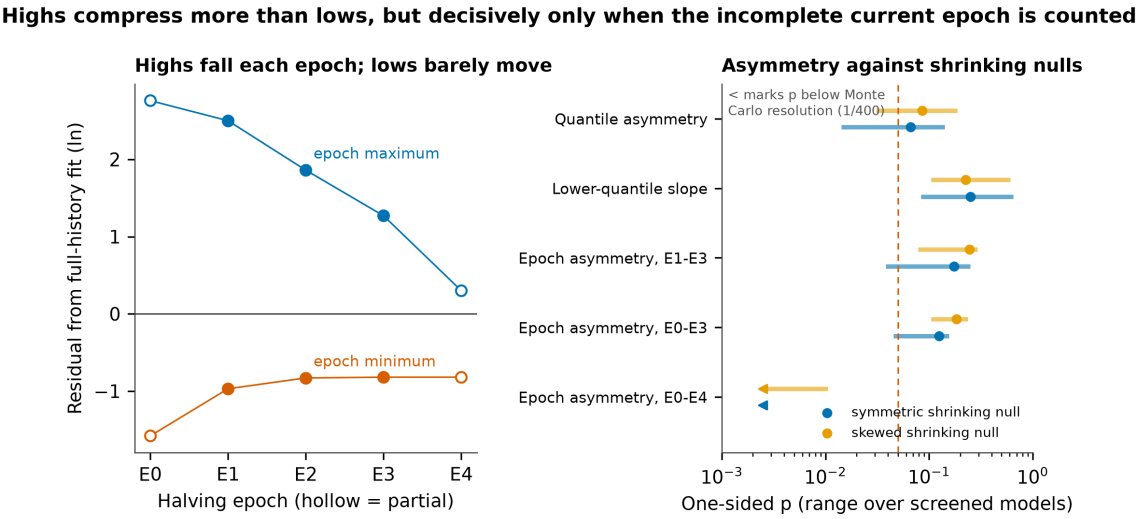


Figure 13: Left: largest and smallest full-history residual in each halving epoch (hollow markers: partial or incomplete epochs). Right: one-sided p-values for top-only compression against symmetric and skewed shrinking nulls; bars show the range across the twelve screened residual models, dots the median (simulation). Dashed line: $p = 0.05$.

8.3 Interpretation

The direction is consistent: every version of the statistic points toward top-only compression. The strength is not. Using only complete epochs, p ranges from 0.04 to 0.28 across screened models and nulls, and the quantile version from 0.015 to 0.18. The result becomes decisive ($p < 0.01$) only when the incomplete current epoch is included. Its maximum so far (0.30) is a lower bound: a rally later in the epoch would raise it and weaken the result. With three complete epochs, the slope of the maxima rests on three points. The lows on their own show nothing unusual ($p = 0.09$–0.62); the asymmetry comes entirely from the highs.

Two features of the design probably make these p-values conservative: the simulated cycle is not tied to halvings, so simulated epoch extremes are noisier than they would be if cycle peaks were phase-locked to the partition; and the screened family may overstate low-frequency variability (Section 7.5). Neither can be quantified without the cycle-validation work that is outside this paper's scope.

We therefore report top-only compression as a descriptive regularity with borderline calibrated support, and as a candidate for re-testing when the current epoch closes. It is consistent with the declining residual amplitude reported by Perrenod [3] and with Banks's finding that the upper envelope scales differently from the lower [4].

9 Keeping Pace Versus Catching Up

Perhaps the clearest result of the study is the distinction between two superficially similar questions.

Question 1. How fast must Bitcoin grow to remain the same distance below the Power-Law trend?

Question 2. How fast must Bitcoin grow to erase its existing deficit and reach the trend?

They are very different problems.

9.1 Keeping pace

The exact next-one-year trend CAGR is 36.63%. Across Bitcoin's full historical rolling one-year return distribution, that hurdle is near the 35th percentile. Approximately 65% of all historical rolling one-year windows exceeded it; 60% of windows beginning on or after January 1, 2016 exceeded it; and 55% of windows beginning on or after January 1, 2020 exceeded it.

Here and in the rest of Sections 9–10, era labels such as "post-2020" are assigned by window start date. This section applies today's fixed hurdle to historical realized returns; Section 10 asks a different question by comparing each historical window with the hurdle that prevailed at that window's own start.

These percentages are not forward-looking probabilities. Rolling windows overlap strongly. They simply show that a one-year return on the order required to keep pace has not been unusual historically.

9.2 Catching up

Closing the existing gap requires much larger returns. Approximate annualized requirements are shown in Table 15.

Table 15: Annualized return required to reach the frozen-fit trend, from August 14, 2026.

Horizon	CAGR required to reach trend
1 year	199.0%
2 years	100.5%
3 years	74.6%
4 years	62.4%
5 years	55.1%

To put those requirements in historical context, we ask what share of post-2020 rolling windows actually produced a realized CAGR at least as high as the catch-up rate shown above. The descriptive shares are approximately 17% for the one-year catch-up rate, 18% for the two-year rate, and about 9.0% for the three-year rate. These horizon-specific occupancy shares need not be monotonic: both the realized-return distribution and the eligible start-date sample change with horizon.

The take-home is simple: keeping pace has been historically common; catching up from the present deficit has been historically much rarer.

The eligible post-2020 start-date span covers only about 1.21 non-overlapping three-year window lengths, so the three-year number should be treated with considerable caution. The post-2020 five-year comparison is suppressed because its eligible start dates represent only about 0.32 independent-window equivalents, too little history to make the percentage useful. These are historical occupancy shares, not forward probabilities.

9.3 Interpretation

The current difficulty is therefore not adequately described as: *"Can Bitcoin still grow at roughly 35–40%?"* The more demanding question is: *"Can Bitcoin grow fast enough, for long enough, to eliminate a roughly 0.78-log accumulated deficit while the trend itself continues rising?"*

That distinction helps reconcile two observations that otherwise appear contradictory. The trend's required growth rate is gradually becoming easier. Yet return to trend can simultaneously become more difficult if a large deficit has already accumulated.

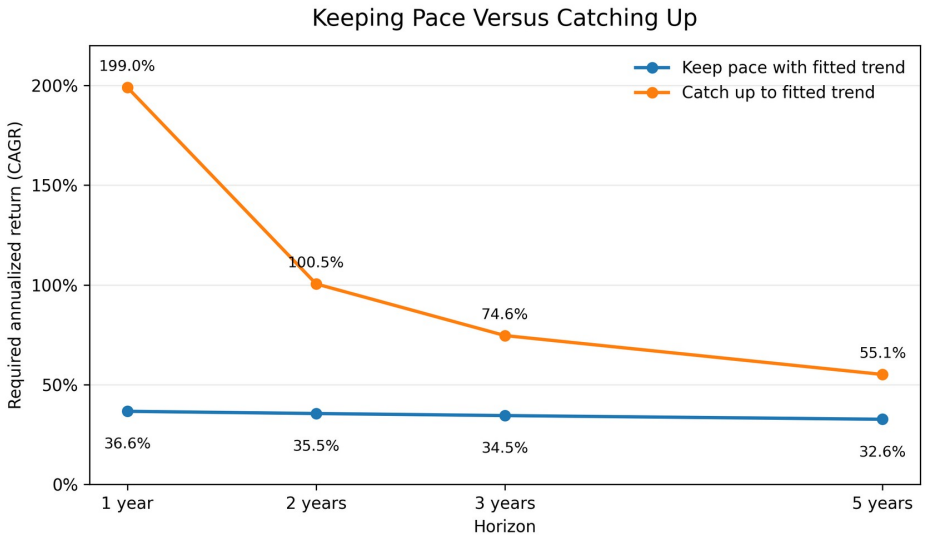


Figure 14: Keeping pace versus catching up. Annualized return required to preserve the current residual (keep pace) or to reach the fitted trajectory by the stated horizon (catch up), frozen at August 14, 2026. (v0.8 Figure 7.)

10 A Separate Open Question: Realized Growth Versus the Contemporaneous Hurdle

The volatility investigation produced another observation that should not be conflated with the volatility-recovery hypothesis. For rolling windows whose start dates are on or after January 1, 2020, we compare realized CAGR with the Power-Law growth hurdle that was actually in force at each window's own start (Table 16).

Table 16: Post-2020 rolling windows: realized CAGR against the hurdle in force at each window's start.

Horizon	Median realized CAGR	Median contemporaneous hurdle	Share beating own hurdle
1 year	∼56%	∼48%	∼52%
2 years	∼42%	∼49%	∼46%
3 years	∼35%	∼49%	∼26%

Thus the one-year comparison remains favorable on median, while the two- and especially three-year

comparisons are less so. This is intentionally different from Section 9, where today's fixed 36.63% keep-pace hurdle is applied to the historical one-year return distribution.

That 26% figure does not mean that Bitcoin has been "above the Power Law only 26% of the time since 2020." It is specifically a comparison of realized three-year growth rates with the growth hurdle that existed at each window's start.

This is potentially important, but the independent sample is extremely small. A similar weak-growth period also appears in an earlier 2018–2022 comparison and subsequently resolved. The correct conclusion is therefore not that a structural maturation squeeze has been identified. It is:

> *Recent Bitcoin has at times failed to generate enough sustained multi-year realized growth to match the contemporaneous Power-Law hurdle. The available independent history is insufficient to determine whether this is cycle-specific noise or the beginning of something structural.*

We leave this observation open rather than adding another research branch to the present study. It is a natural candidate for future work as additional independent multi-year data accumulate, and other researchers may wish to test it with different methods.

11 Discussion

11.1 What survives the stress tests

Several properties of the empirical relationship remain intact.

First, the regression itself is highly reproducible. Our estimate at the Santostasi–Perrenod cutoff is essentially identical to theirs.

Second, the exponent is robust to ordinary changes in sampling frequency.

Third, apparent structural breaks lose evidentiary force when the detector is applied to an appropriate persistent-residual null. Both the number and magnitude of detected breaks are ordinary under that null, and version 0.9 finds the same under every screened model of a broader residual-model family.

Fourth, the large movements in rolling-window and post-2017 estimates do not, on the evidence available, establish a decline in the exponent. They follow from the short span of $\ln t$ covered by a few calendar years and from the persistence of the residuals, and a constant exponent produces movements of the same size under the screened residual models.

Fifth, the current model is not simply eliminating its own error through rapid refitting. During the current episode, approximately 89% of the original dollar deficit and 92% of the original log deficit remain after refitting.

Sixth, the trend's required percentage growth falls with time rather than rising indefinitely.

None of these findings proves that the model will continue to describe Bitcoin indefinitely. What survives is the empirical description under the tests completed here — not proof of a law of nature. The tests do, however, weaken or rule out several narrower failure explanations within the scopes tested.

11.2 What does not survive

The claim that the expanding Bitcoin exponent has converged to a sharply known constant does not survive robust uncertainty analysis. The current estimate is far less precise than naive OLS suggests, and simulation under the screened residual models indicates that even the conventional HAC interval understates the uncertainty under most of them.

The converse claim also does not survive: that the rolling five-year or post-2017 estimates show the exponent has structurally declined. Those estimates are not unusual under a constant exponent with persistent residuals, except under the least persistent corner of the residual-model family.

Neither does the inference that non-rejection proves constancy. The tests available have low power: under critical values valid across the screened family, none detects a three-unit decline more than about one time in five.

Similarly, our exact residual series does not yield a clean, unanimous stationary-residual result across ADF, Phillips–Perron, and residual-based tests. The v0.8 Phillips–Perron value of -2.380 is withdrawn; the reproducible values (-3.04 and -3.31) do not reject at residual-based 5% critical values either, and no statistic rejects against a unit root with Bitcoin's declining volatility.

The claim that lower volatility itself explains slower recovery also does not survive decomposition of the mechanism.

And the idea that the current raw deficit is historically unprecedented is false. The latter distinction matters: the present episode is unusual relative to the modern volatility regime, not because its raw log gap exceeds all historical precedents.

11.3 Evidence status of the principal claims

Table 17 classifies the principal claims by the kind of evidence behind them. "Deterministic" results are computed directly from the frozen series and reproduce exactly. "Analytic" results follow mathematically from stated assumptions. "Simulation" results depend on simulated residual processes and hold only to the extent that those processes resemble Bitcoin's; none of them is an independently verified empirical fact.

Table 17: Evidence status of the principal claims.

Claim	Evidence type	Status and conditions
Baseline fit ($B = 5.6509$, $R^2 = 0.961$) and expanding-window estimates	Deterministic	Reproduced exactly; conditional on the genesis time origin
Rolling and post-2017 estimates move by large amounts; the 2025 five-year decline came from 2020 data leaving the window	Deterministic, with analytic explanation	Reproduced exactly
Those movements are compatible with a constant exponent	Simulation	Principal statistic: all 12 screened models ($p = 0.14$–0.35); secondary battery: 11 of 12 (adjusted $p \geq 0.05$); conditional on the declared family and screen
B is constant	—	Not established: low power
B has declined structurally	—	Not established
Conventional HAC intervals for B under-cover	Simulation	Under 10 of 12 screened models; conditional on the family
BIC break detector is uninformative	Simulation	Under the v0.8 AR(1) null and all screened models; conditional on detector and null
Residual unit root versus stationarity	Deterministic statistics, simulated critical values	Undetermined; tests cannot discriminate
Corrected Phillips–Perron statistic	Deterministic (`arch` 8.0.0)	-3.04 (lag 11), -3.31 (automatic); v0.8 value not reproduced
Equal-standardized-gap scale invariance	Analytic (illustrated by simulation)	Conditional on a stationary AR(1); not evidence of mean reversion
A four-year residual cycle	Imposed in the candidate family	Models containing it pass the screen; not independently identified or tested in this paper
Current episode: mid-field in raw depth, deepest in local-sigma depth	Deterministic	Conditional on the 60-day episode rule and four-year local window
Refitting absorbs a small share of the current deficit	Deterministic	Limited to the current episode
Declining volatility does not explain slower recovery (H1)	Analytic and descriptive	Persistence channel not excluded
Keeping pace versus catching up	Deterministic arithmetic; descriptive occupancy shares	Shares are not forward probabilities
Highs compressing faster than lows	Deterministic (descriptive); simulation (test)	Borderline with complete epochs; decisive only with the incomplete current epoch

11.4 What is genuinely unusual about the current episode

The current episode is unusual not because Bitcoin has traveled farther below the trend in absolute log distance than it did in 2022. It has not. It is unusual because:

1. the same raw gap has occurred in a much lower residual-volatility environment;
2. the recent residual distribution itself is centered materially below the trend;

3. the episode became progressively weaker through June 2026; and
4. the current price must both keep pace with a rising trend and close a large accumulated deficit.

These points are distinct. The first creates the extreme local-sigma trend-depth reading. The second explains why that reading looks much less extreme after local de-meaning. The third documents the path into the current low. The fourth explains why a declining future trend hurdle does not automatically make near-term recovery easy.

11.5 Current episode as a live test

Prior work has already described the 2025–2026 period as a live test of the Bitcoin Power Law. We agree with the framing but treat it quantitatively rather than rhetorically. At the August 14, 2026 freeze:

- price is approximately \$63.0k;
- fitted trend is approximately \$137.9k — the model-implied Power-Law trend level;
- raw residual is -0.7833;
- trend-centered local-sigma depth is -2.521; and
- locally de-meaned z is -1.687.

The largest historical cumulative below-trend episode had area, not point depth, of approximately 537.6 log-days. The current episode has accumulated approximately 137 log-days through the cutoff, well below that completed historical maximum in cumulative area. This does not contradict its unusually large modern-standardized depth: one metric measures how deep the episode gets at a point; the other measures how much below-trend deficit accumulates over time.

A duration beyond approximately March 22, 2028 without a trend crossing would exceed the longest prior episode under the current episode definition. These thresholds are not forecasts and should not be treated as magical falsification points. They are pre-specified empirical markers that would make the current episode increasingly difficult to describe as ordinary historical persistence.

11.6 Falsifiability matters more than narrative

The Power Law is easiest to misuse when every new outcome is made consistent with it after the fact. That is why the current episode is valuable analytically. A useful empirical framework should state in advance what future observations would reduce confidence in it. Falsifiability does not mean hoping the model fails; it means refusing to move the goalposts after unfavorable evidence arrives.

The monitoring markers fixed before publication of v0.8, reproduced here unchanged, include:

- no trend crossing beyond roughly March 22, 2028, corresponding to a duration beyond the prior $\sim$870-day maximum under this episode rule;
- cumulative below-trend area above approximately 537.6 log-days;
- a trailing four-year residual mean below approximately -0.40 and still falling;
- continued residual deterioration combined with a materially larger downward movement in B, beyond the scale already observed, while residuals remain negative; and
- persistent multi-year realized growth below the contemporaneous hurdle across genuinely new independent windows.

Not every future warning can be reduced to a single timeless cutoff: robust uncertainty around B, for example, will itself evolve as new data arrive. The point is to specify the form of adverse evidence in

advance rather than invent a new rescue narrative after it occurs. The model should not be "rescued" by inventing explanations only after those events occur.

One further question lies outside the present study: whether observed residual deficits can be related to economically plausible capital flows. Future work could examine the scale of net allocation or systematic rebalancing demand associated with closing a given Power-Law residual, conditional on market capitalization, liquidity, and the size of the addressable capital pool.

12 Limitations

12.1 Time-origin dependence

The fitted exponent depends materially on the time origin. Baquero and Menezes [2] report nearly a threefold change in the fitted exponent across an additive time-shift sweep, although their information criterion is minimized at the unshifted genesis convention. We use Bitcoin genesis because it defines the model being studied, not because the exponent is mathematically invariant to alternatives. Any structural interpretation of the numerical value of B must acknowledge this dependence.

12.2 Persistent residuals

Residual autocorrelation near 0.997 creates severe inferential complications. HAC estimates and block bootstraps help, but no single bandwidth or block length is uniquely correct, and simulation under the screened residual models (Section 6.2) shows that the conventional intervals under-cover under most of them. The uncertainty estimates reported here are therefore better viewed as a robust range, indexed by the residual model, than as a precise confidence statement.

12.3 Stationarity ambiguity

Our residual diagnostics are mixed. Against residual-based critical values, the Phillips–Perron statistics do not reject a unit root and ADF rejection depends on lag specification; against a unit-root null with Bitcoin's declining volatility, no statistic rejects. This limits strong claims that deviations are governed by a stable stationary mean-reverting process.

It also qualifies the AR(1) mechanism illustration in Section 6.9. That exercise assumes a stationary AR(1) with $|\rho| < 1$ so that a stable process variance exists and standardized recovery can be compared across scales. We use it to isolate the volatility-scale channel, not to assert that Bitcoin residuals have been proven to follow that process. If the true residual dynamics are unit-root-like or otherwise nonstationary, the mean-reversion interpretation itself becomes weaker; the AR(1) result remains a conditional statement about the mechanism tested.

12.4 Small number of completed episodes

Only nine completed recovery episodes are available, with still fewer possessing full local-volatility history. Episode-level correlation statistics are exploratory. Put simply, nine historical episodes are not enough to estimate a precise population relationship between volatility, depth, and recovery time. These correlations are descriptive rather than precise population inferences with conventional statistical significance.

12.5 Overlapping return windows

Rolling one-, two-, three-, and five-year return windows overlap substantially within each horizon. For example, two consecutive three-year windows that start one day apart share almost all of their observations.

They are therefore not independent three-year experiments simply because a computer can calculate thousands of them. Percentages of windows exceeding a hurdle measure historical occupancy, not independent success probabilities. This is especially important for post-2020 multi-year comparisons. The same overlap applies to rolling exponent estimates (Section 7), whose successive values are not independent observations of the exponent.

12.6 Structural-break detector limits

The segmentation algorithm was capped at four breaks and selected that boundary. Our conclusion is therefore not that the optimal number of breaks equals four. Rather, the detector itself is shown to produce similar behavior under the constant-slope null and is consequently uninformative for the structural question in its current form. That conclusion is conditional on the detector and on the simulated residual processes used as controls. Higher break caps were outside the scope of this study.

12.7 Simulation scope

The AR(1) simulations of Section 6.9 are mechanism illustrations. They do not reproduce Bitcoin's heavy tails, asymmetric returns, changing persistence, cycle structure, changing drift, or market-regime shifts. Their useful result is analytic: scale alone cannot change standardized first-passage behavior when persistence is fixed.

The residual-model family of Section 7.4 is broader but still limited. Its innovations are Gaussian; heavy tails and volatility clustering within years are not modelled, and right skew is imposed only in separate checks (Sections 7.5 and 8). It matches the direction but not the exact size of the decline in daily volatility (simulated about 0.31 against an observed 0.28 for the ratio of late to early daily-return volatility).

12.8 Declared residual-model family and the four-year cycle (new)

Every simulation-based probability in Sections 6.2–6.4, 7 and 8 is conditional on the declared family and on the screen used to select models from it. The family imposes a stochastic cycle with a period fixed in advance at about four years (3.5 and 4.5 years in sensitivity runs), and two of the six screening features target multi-year periodicity. That every screened model contains the cycle is therefore a property of the design, not an independent finding. Other residual structures, including log-periodic components, other long-memory processes and regime-switching models, were not examined, and their implications for the calibrated p-values, power and coverage are unknown.

The screen and the tests also use the same history. If the exponent did decline, part of that decline would appear in the constant-B residuals as slow low-frequency variation and could favor models with more of it. A slow change in the trend and persistent residual variation cannot be fully separated in one sixteen-year history.

12.9 Post-hoc selection and multiplicity (new)

The statistics examined in Section 7, including the principal reported curvature test, were chosen after the historical series had been studied. The Westfall–Young adjustment accounts for multiplicity within the six-statistic battery but not for the choice of the battery itself, and the curvature test is not prospectively preregistered. The selection-aware minimum of the rolling path addresses one specific form of after-the-fact selection. The reported p-values should be read as calibrated descriptions under the stated family rather than as confirmatory tests.

12.10 No independent forecasting study

This paper studies empirical stability, residual behavior, and recovery conditions. It does not conduct a new out-of-sample forecasting tournament. Baquero and Menezes [2, 10] already provide a more direct analysis of that question.

12.11 No claim of causal mechanism

The analysis is primarily price-based. We do not estimate the network-adoption and generalized Metcalfe components required to establish the causal or mechanistic interpretation proposed in other work. In other words, reproducing and stress-testing the price–age relationship does not tell us whether network effects are the causal reason that relationship exists.

12.12 Local-standardization window

The four-year trailing window is the local-standardization convention used in this study. The headline raw-versus-local comparison is therefore conditional on that window choice. We did not run a new sensitivity sweep of episode rankings over alternative two-, four-, or six-year local-volatility denominators after the research freeze; that is a natural subject for future work rather than a result implied here.

12.13 Residual-model alternatives

The break-null and recovery simulations treat residual structure as persistent stochastic variation, using AR-type dependence as the mechanism under study. Other work models Bitcoin's deviations with deterministic or quasi-deterministic log-periodic components [7, 8]. We do not adjudicate between those residual specifications in this paper. Our conclusions about break-detector behavior, AR(1) scale invariance and the calibrated exponent tests are conditional on the residual models actually tested.

12.14 Episode-definition sensitivity

The episode inventory and all episode-derived rankings, recovery comparisons, and forward monitoring markers are conditional on the stated 60-day minimum-duration rule and the strict-crossing/bridge conventions in Section 5.5. The 3-day and 7-day bridge tolerances were used as robustness checks for short interruptions around zero, but this study did not sweep alternative minimum-duration thresholds such as 30 or 90 days. Accordingly, quantities such as the episode count, historical severity ranks, longest-prior-episode duration, and cumulative-area benchmark should be read as conditional on the canonical episode definition rather than invariant features of the data.

12.15 Data provenance and reproducibility scope (new)

The source composition of the frozen series is inferred rather than proven row by row (Section 3), and the provider transition falls inside the current episode. The licensed price observations are not redistributed, so exact reproduction requires the frozen file or an independently obtained series, which will generally differ slightly. Several secondary v0.8 diagnostics were produced by scratch code that was not preserved; the repository identifies which results it reproduces exactly, which it reproduces by method only, and which it does not reconstruct (see Data and code availability). The v0.8 Phillips–Perron statistic could not be reconciled with any standard implementation and has been replaced.

12.16 Material not re-examined

Baquero and Menezes' replication package [10] could not be retrieved for this revision, so their low-frequency test statistic was not re-implemented; our family is an adaptation of their design, checked against their published figures (Appendix D).

13 Conclusion

Bitcoin's historical price remains descriptively close to a power-law function of its age under the conventional genesis-date time origin: the log–log regression has $R^2 = 0.961$, while one full-history residual standard deviation still corresponds to roughly a factor-of-two deviation in price. That descriptive statement survives. Several stronger statements do not.

The fitted exponent is less precisely known than naive OLS suggests, and, under most of the residual models we simulated, less precisely known than conventional robust intervals suggest. Its apparent expanding-window stabilization is partly a mechanical consequence of accumulating data. The post-2021 decline in the expanding estimate of B is real numerically, but it does not on its own establish structural decay. The longer comparison is also less dramatic: the 2026 estimate remains slightly above the 2016 estimate, reinforcing that a multi-year downward run should not automatically be read as permanent exponent decay.

The same caution applies with more force to short windows. Rolling five-year and post-2017 estimates have moved by several units, and the five-year estimate fell to about 3.26 during 2025. Those movements follow from the narrow span of $\ln t$ that a few calendar years cover and from the persistence of the residuals; the 2025 fall came from 2020 observations leaving the window. Under the screened residual models, a constant exponent produces movements of this size routinely. But the tests are weak: with critical values valid across the screened models, none detects even a three-unit decline more than about one time in five. Neither the non-rejection nor the strong descriptive fit establishes that B is constant. With sixteen years of data, the question is open.

Statistical break detection likewise requires caution. A detector that appears to find four historical breaks also finds four breaks almost universally in constant-exponent simulations with persistent residuals, and the magnitudes of the real breaks are ordinary under that null. Unit-root tests on the residuals cannot separate a highly persistent stationary residual from a unit root, particularly once Bitcoin's declining volatility is taken into account.

The current 2025–2026 below-trend episode is not unprecedented in raw distance. Its maximum raw residual is effectively the same as the 2022 minimum. What has changed is the environment around that gap. Residual volatility is substantially lower, and the recent residual distribution itself is centered materially below zero. The current episode is therefore extreme when measured as distance from the Power-Law trend relative to modern residual fluctuation size, but only marginally more extreme than 2022 under a conventional locally de-meaned z-score.

This distinction matters. A trend-centered standardized measure asks whether price is far from the model. A de-meaned measure asks whether price is far from recent experience. The current market is more unusual under the first question than the second because recent experience has itself shifted below the long-run trend.

The hypothesis that declining volatility causes slower recovery is not supported. Lower variance makes a fixed raw gap larger in standardized units, but it cannot alter equal-standardized-gap recovery under fixed-persistence AR(1) dynamics. In plainer terms: if two episodes start equally far from trend relative to the volatility of their own eras, simply lowering the volatility scale does not make the lower-volatility episode recover more slowly. Persistence itself does not exhibit a sufficiently stable maturation pattern in our data to restore the hypothesis. This is a conditional mathematical result, not evidence that Bitcoin's residuals mean-revert.

The trend's own growth requirement is also declining. Bitcoin does not need an ever-increasing percentage

growth rate to remain on a t^B trajectory. But that fact should not be mistaken for an easy recovery path. At the August 14, 2026 data cutoff, maintaining the current distance below the model-implied trend would require approximately 36.6% Bitcoin price appreciation over the following year. Fully closing the existing deficit would require approximately 199% price appreciation over one year, 100.5% CAGR over two years, or 74.6% CAGR over three years. These are model-implied growth requirements, not forecasts. The accumulated deficit — how far price currently sits below the model-implied trend level — not simply the slope of the trend, is the binding constraint.

The resulting conclusion is deliberately limited:

> *Bitcoin's historical price continues to exhibit a strong descriptive Power-Law relationship with age under the specified time-origin convention. The completed robustness tests do not provide compelling evidence of structural failure or of a structural decline in the exponent, but they also do not establish an immutable law or a constant exponent. The current below-trend episode remains a meaningful live test: raw underperformance is historically familiar, modern trend-centered depth is unusually large, the recent residual center is persistently negative, and whether future realized growth is sufficient to close the deficit remains unresolved.*

That unresolved part is not a defect in the analysis. It is the question future data will answer.

Data and code availability

The frozen 5,873-row BTC/USD series used in this paper is derived from licensed provider data and is **not redistributed**. The same applies to daily residual or fitted-value tables, figure-source tables containing daily observations, per-row hashes and any other artifact that would effectively reconstruct the provider series. Version 0.8 stated that the frozen dataset would be released with the final publication version; after review of upstream redistribution restrictions that plan was withdrawn, and this statement supersedes it.

The public repository, `https://github.com/fadethecrowd/bitcoin-power-law-paper`, publishes the reproduction code, statistical conventions, aggregate verification targets, the input schema, provenance documentation, and the whole-file SHA-256 of the exact frozen CSV:

`ada93367320ad9c45d713650377ee683cf0b58d1ff8fb290fb72cd8b3168886f`

A reader who holds that exact file can confirm byte identity; a reader who obtains Bitcoin history independently from a lawfully usable source can rerun every calculation in methodological form, with results that will be close but not identical.

The v0.9 robustness analyses (Sections 6.2–6.4, 7 and 8) are produced by a separate reproducibility package added to the repository with this version. It contains the scripts, dependency versions (Python 3.13; `numpy` 2.5.3, `scipy` 1.18.1, `pandas` 3.0.5, `statsmodels` 0.15.0, `arch` 8.0.0, `matplotlib` 3.11.2), every simulation setting and seed, aggregate outputs and the figures. All simulation seeds derive from a master seed, 20261025, by stage and model index; with the frozen file, every simulation reproduces bit for bit. Outputs are aggregate statistics; no daily series are written. Point values are limited to those printed in this paper: the halving-epoch residual extremes, for example, are given to two decimals and without dates, so they cannot be matched to an individual day's close. Appendix E maps each result to the script that produces it.

The repository distinguishes three levels of reproduction: exact frozen-file identity, deterministic statistical reproduction, and stochastic-method reproduction. The original random seeds of the v0.8 bootstrap and break-null runs were not preserved, so those results are reproduced by method with a new publication seed (20261001), not draw for draw. A small number of secondary v0.8 diagnostics, listed in the repository's reproducibility-scope document, were produced by scratch code that was not preserved and are not reconstructed.

The v0.8 paper is archived at Zenodo, https://doi.org/10.5281/zenodo.22966136 (all versions: https://doi.org/10.5281/zenodo.22966135).

Authorship and AI assistance

Author: Maxwell Koenig. The author directed the research, reviewed the research questions, methodology, results and interpretations, and is responsible for the paper's content, including any errors.

AI assistance: This work was developed with substantial help from two large language models. **ChatGPT (OpenAI)** served as a primary thinking partner in developing and critically testing research questions and interpretations, and assisted with manuscript review and independent checks of selected statistical results and reproducibility materials. **Claude (Anthropic)** carried out much of the analytical work, including regressions, simulations, robustness checks, figure verification, literature review, preparation of reproducibility code and materials, and manuscript revisions. These contributions are acknowledged as AI assistance, not as additional human authors or Zenodo creators.

Version history

Version 0.9 (October 2026) revises v0.8 (September 2026) on the same frozen sample. The substantive changes are: new Sections 7 and 8; simulation-based coverage of the intervals for B (Section 6.2); corrected Phillips–Perron statistics and residual-based critical values (Section 6.3); calibration of the break detector under a broader residual-model family (Section 6.4); revised abstract, discussion, limitations, conclusion and data-availability statement; and new Appendices C–E. The v0.8 forward-monitoring markers, episode definitions, time origin and model specification are unchanged. Figures have been renumbered; v0.8 figures are retained with their original content. A separate change log lists each modification.

References

[1] Santostasi, G., & Perrenod, S. (2026). A mechanistic derivation of the Bitcoin price power law: Network adoption dynamics and generalised Metcalfe scaling. *Nonlinear Science*, 8, 100172. https://doi.org/10.1016/j.nls.2026.100172. Preprint: https://doi.org/10.5281/zenodo.19387099.

[2] Baquero, C., & Menezes, R. (2026). Bitcoin's Power Law: Weak Structure, Strong Forecasts. arXiv:2605.21316. https://arxiv.org/abs/2605.21316.

[3] Perrenod, S. (2026, June 5). Bitcoin's Power Law Passes a Major Statistical Test. *Money or Debt Newsletter*. https://stephenperrenod.substack.com/p/bitcoins-power-law-passes-a-major.

[4] Banks, P. J. (2022, May 18). The Bitcoin Power Law Model. *Parker's Blog*. https://pbanks.net/blog/the-bitcoin-power-law-model.html.

[5] Pessa, A. A. B., Perc, M., & Ribeiro, H. V. (2023). Age and market capitalization drive large price variations of cryptocurrencies. *Scientific Reports*, 13, 3351. https://doi.org/10.1038/s41598-023-30431-3.

[6] CryptoAdvisor. (2026, February 8). Mean Reversion: The "Gravity" Behind Bitcoin's Price. *Bitcoin Trading Ideas, Analysis* (practitioner blog). https://btctrading.blog/2026/02/08/4689/.

[7] Wheatley, S., Sornette, D., Huber, T., Reppen, M., & Gantner, R. N. (2019). Are Bitcoin bubbles predictable? Combining a generalized Metcalfe's Law and the Log-Periodic Power Law Singularity model. *Royal Society Open Science*, 6(6), 180538. https://doi.org/10.1098/rsos.180538.

[8] Perrenod, S. (2026, March 23). Bitcoin's Scaling Law: Power Laws, Log Periodicity, and a Hidden Coupling. *Money or Debt Newsletter*. https://stephenperrenod.substack.com/p/bitcoins-scaling-law-power-laws-log.

[9] wineLightning, & Claude AI. (2026, July 9). Bitcoin's Price Power Law Decomposed: Epidemic Speed Times Wealth Inequality (Version v2). Zenodo. Preprint; companion code and data are publicly available. https://doi.org/10.5281/zenodo.21270485.

[10] Baquero, C., & Menezes, R. (2026, October). Deceleration without identification: Testing a power-law trend in Bitcoin prices under persistent noise. Working paper, version 1 (not peer reviewed). Zenodo. https://doi.org/10.5281/zenodo.23218857. Replication package: https://doi.org/10.5281/zenodo.23218897.

[11] Santostasi, G., & Perrenod, S. (2026, October 2). A Clauset-in-spirit goodness-of-fit test for a power law in time: Application to Bitcoin, 2010–2026. Zenodo (not peer reviewed). https://doi.org/10.5281/zenodo.23105856.

[12] Clauset, A., Shalizi, C. R., & Newman, M. E. J. (2009). Power-law distributions in empirical data. *SIAM Review*, 51(4), 661–703.

[13] Durlauf, S. N., & Phillips, P. C. B. (1988). Trends versus random walks in time series analysis. *Econometrica*, 56(6), 1333–1354.

[14] Phillips, P. C. B. (2007). Regression with slowly varying regressors and nonlinear trends. *Econometric Theory*, 23(4), 557–614.

[15] Nelson, C. R., & Kang, H. (1981). Spurious periodicity in inappropriately detrended time series. *Econometrica*, 49(3), 741–751.

[16] Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.

[17] Künsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics*, 17(3), 1217–1241.

[18] Dickey, D. A., & Fuller, W. A. (1979). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association*, 74(366), 427–431.

[19] Phillips, P. C. B., & Perron, P. (1988). Testing for a unit root in time series regression. *Biometrika*, 75(2), 335–346.

[20] Engle, R. F., & Granger, C. W. J. (1987). Co-integration and error correction: Representation, estimation, and testing. *Econometrica*, 55(2), 251–276.

[21] Phillips, P. C. B., & Ouliaris, S. (1990). Asymptotic properties of residual based tests for cointegration. *Econometrica*, 58(1), 165–193.

[22] MacKinnon, J. G. (2010). Critical values for cointegration tests. Queen's Economics Department Working Paper No. 1227, Queen's University.

[23] Schwarz, G. (1978). Estimating the dimension of a model. *The Annals of Statistics*, 6(2), 461–464.

[24] Harvey, A. C. (1989). *Forecasting, Structural Time Series Models and the Kalman Filter*. Cambridge University Press.

[25] Westfall, P. H., & Young, S. S. (1993). *Resampling-Based Multiple Testing: Examples and Methods for p-Value Adjustment*. Wiley.

[26] Koenker, R., & Bassett, G. (1978). Regression quantiles. *Econometrica*, 46(1), 33–50.

[27] Seabold, S., & Perktold, J. (2010). statsmodels: Econometric and statistical modeling with Python. In *Proceedings of the 9th Python in Science Conference*, 92–96.

[28] Sheppard, K., et al. arch: ARCH models and unit-root tests in Python, version 8.0.0 [software]. https://github.com/bashtage/arch.

A Canonical Numerical Snapshot and Conditional AR(1) Illustration

Table 18 reproduces the v0.8 audited snapshot unchanged. Table 19 adds the principal v0.9 quantities.

Table 18: Canonical numerical snapshot (v0.8 audited values, unchanged).

Quantity	Audited value
Observations	5,873
Sample	2010-07-17 to 2026-08-14
A	4.156832×10^{-17}
B	5.650938
R^2	0.961281
Full-history residual SD	0.697088
Residual lag-1 autocorrelation	0.996985
HAC SE of B, lag 365	0.18916
365-day block-bootstrap SD of B	0.2401
365-day block-bootstrap 95% interval	[5.0466, 5.9960]
Current completed close	\$62,996.40
Current trend	∼\$137.9k
Current raw residual	−0.7833
Current local-sigma trend depth	−2.521
Current local de-meaned z	−1.687
Current RMS trend depth	−1.936
Current trailing-4y residual mean	−0.2590
Current trailing-4y residual SD	0.3107
Episode maximum raw depth	−0.8165
Maximum-depth date	2026-06-30
Episode-minimum local-sigma trend depth	−2.732
Episode-minimum de-meaned z	−1.907
Current cumulative below-trend area	∼137 log-days
Largest prior episode area	537.6033 log-days
Arithmetic equivalent of instantaneous annualized log-growth rate	37.84%
Exact one-year trend CAGR	36.63%
One-year catch-up CAGR	199.0%
Two-year catch-up CAGR	100.5%
Three-year catch-up CAGR	74.6%
Refit absorption, log space	7.84%
Refit absorption, price space	11.26%

Inferred source of the current completed close: CoinGecko (coingecko.com).

Table 19: Principal v0.9 quantities.

Quantity	Value	Type
Post-2017 B (2017-01-01 to 2026-08-14)	4.8645	Deterministic
Rolling five-year B at end-2024 / end-2025 / 2026-08-14	4.2004 / 3.2648 / 4.1151	Deterministic
2025 change in rolling five-year B: leaving / entering	−2.03 / +1.09	Deterministic
Curvature statistic (implied change in local exponent)	−1.758	Deterministic
Phillips–Perron Z_t, lag 11 / automatic (34)	−3.04 / −3.31	Deterministic (`arch`)
Screened models in declared family	12 of 48	Simulation
Principal statistic (curvature) p, range over screened models	0.14–0.35	Simulation
Secondary battery, adjusted p, range over screened models	0.011–0.56	Simulation
Family-wide power at $D = 3$, best statistic	0.18	Simulation
95% interval for D by curvature inversion, best-fitting model	[−2.5, 5.9]	Simulation
NW-365 coverage of nominal 95% interval, screened models	0.29–0.99	Simulation

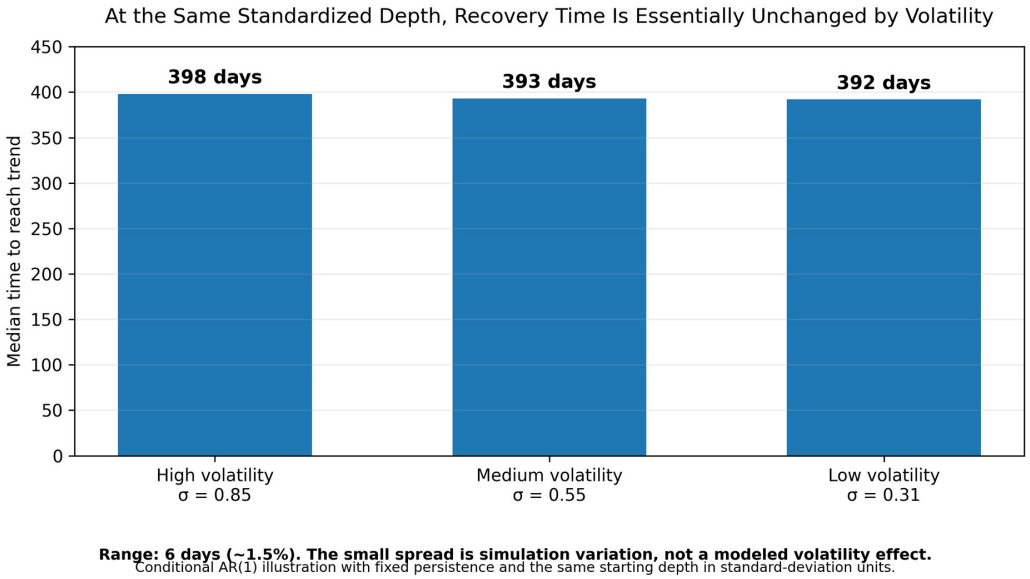


Figure 15: Equal-standardized-gap recovery is scale-invariant under fixed-persistence AR(1). Median simulated time to reach trend from the same standardized depth at three volatility scales. A conditional illustration of an analytic result; it is not evidence that Bitcoin's residuals follow a stationary AR(1). (v0.8 Appendix Figure A1.)

B Claims Deliberately Not Made

This paper does not claim that:

- the numerical exponent is invariant to time-origin choice;
- a high R^2 proves a structural law;
- residuals cleanly reject a unit root under every specification;
- the residual process has been proven to be a stationary AR(1);
- the break analysis proves that structural change is impossible;
- the local-sigma trend-depth statistic is a Gaussian z-score;
- the four-year local-standardization ranking is invariant to other trailing-window choices;
- declining volatility has been proven incapable of affecting recovery through every possible mechanism;
- historical rolling-window occupancy rates are forward probabilities;
- recovery from prior below-trend episodes implies recovery from the current one;
- the current-episode refit decomposition generalizes to all historical periods;
- Santostasi and Perrenod's mechanistic derivation has been validated by reproducing their regression;
- the Power Law guarantees any future Bitcoin price;
- *(v0.9)* the exponent is constant, or that it has declined structurally;
- *(v0.9)* a non-rejection by a low-power test is evidence that the exponent is constant;
- *(v0.9)* Bitcoin's residuals contain a four-year cycle: the cycle is part of the simulation design, not a finding;
- *(v0.9)* the principal reported curvature test was specified before the data were examined; or

- *(v0.9)* simulation-based p-values, power and coverage rates hold outside the declared residual-model family.

C Declared Residual-Model Family: Settings and Seeds

Table 20: Simulation settings for the declared residual-model family.

Setting	Value
Persistent component	AR(1), $\rho \in \{0.990, 0.995, 0.997, 0.998, 0.999\}$; random walk; fractional integration $d \in \{0.6, 0.9\}$; started at zero at genesis
Stochastic cycle	Harvey trigonometric cycle; period 4 years (3.5 and 4.5 in sensitivity runs); correlation envelope halves after 3 periods; stationary start
Cycle variance share w	0, 0.4, 0.8
Volatility shape λ	0 and 1 (0.5 and 1.25 in sensitivity runs); shocks scaled by the observed annual SD of daily log returns raised to λ
Scale	Each path rescaled so that its constant-B OLS residual SD equals 0.697088
True exponent	$B = 5.650938$; genesis origin 2009-01-03; observed daily calendar
Innovations	Gaussian; right skew imposed by rank mapping in separate checks
Screening features	Residual autocorrelation at lags 1, 365, 1,461 days; zero crossings; SD ratio, last five years over first five; share of variance at 3–5-year periods
Screening rule	Mahalanobis $p \geq 0.05$ on the six features jointly and every feature $p \geq 0.01$
Histories per model	Screening and calibration 1,000; power 1,000 per D (common random numbers); interval coverage and unit-root rates 300; asymmetry 400; unit-root critical values 2,000
Rolling windows	$(e - L$ years, $e]$ for $L = 4, 5, 6, 8, 10$; evaluated at month ends
Multiplicity adjustment	Westfall–Young min-p over d17, curv, mean5, r5_2025, rmin5, exp_drop
Seeds	`SeedSequence([20261025, stage, model index])`; stages: screen 1, calibrate 2, power 3, coverage 4, asymmetry 5, sensitivity 6, skewed battery 7, critical values 8

D Checks of the Simulator Against Baquero and Menezes

Table 21 compares this paper's simulator with benchmark figures published by Baquero and Menezes [10]. The first two rows compare with their Figure 1b, which uses weekly prices from 2011; the remaining rows compare with their Appendix G, which uses simulated daily histories (500 per deviation model, true exponent 5.7). Our simulator works on the daily calendar from 2010, so close rather than identical values are expected.

Table 21: Simulator checks against published Baquero–Menezes benchmarks.

Check	Baquero & Menezes	This paper
Constant drift + random walk: median fitted B (90% range)	6.1 (3.4–9.2)	6.35 (3.09–9.62)
Same: median R^2 / share above Bitcoin's R^2	0.897 / 0.14	0.888 / 0.106
ADF rejection, random walk with declining volatility	0.44	0.40
ADF rejection, random walk, constant volatility	0.18	0.20
ADF rejection, random walk with cycle and declining volatility	0.20	0.21
Newey–West (four-year lag) coverage, random walk	0.34–0.35	0.31–0.32

E Reproducibility Map

Table 22: Where each v0.9 result is produced. Scripts are in `v09_robustness/scripts/` and outputs in `v09_robustness/outputs/` of the repository; outputs are aggregate. The last row refers to the repository's top-level `scripts/` and `tests/`.

Result	Script and output
Baseline, expanding, rolling, start-date and post-2017 estimates; 2025 decomposition; ADF and PP statistics; quantile slopes; epoch extremes	`01_reproduce.py` → `reproduction.json`
Simulation of the 48-model family	`02_simulate_nulls.py` → `cache/` (simulated histories only)
Screening, calibrated p-values, Westfall–Young adjustment, break-detector null (Tables 10, 11)	`03_calibrate.py` → `screening.csv`, `calibration.csv`, `calibration_summary.csv`, `maxT.csv`, `breaks_null.csv`
Power and false-positive rates (Table 12)	`04_power.py` → `power_curves.csv`, `size_table.csv`
Interval coverage, unit-root rejection rates, AR(1) illustration (Table 4)	`05_hac_unitroot.py` → `hac_coverage.csv`, `unitroot.csv`, `scale_invariance.json`
Compression tests (Table 14)	`06_asymmetry.py` → `asymmetry.csv`
Figures 3 and 8–13	`07_figures.py` → `figures/`
Baquero–Menezes check (Table 21)	`08_bm_crosscheck.py` → `bm_crosscheck.json`
Sensitivity variants	`09_sensitivity.py` → `sensitivity.csv`
Skewed-null battery (Table 10)	`10_skew_battery.py` → `skew_battery.csv`
Unit-root critical values (Table 5)	`11_unitroot_critical_values.py` → `unitroot_critical_values.csv`
v0.8 deterministic results	`scripts/reproduce_v08.py`, checked by `tests/test_reproduction.py` against `expected/v0.8_expected_results.json`